AS3811

FINAL YEAR PROJECT REPORT

Scalable Conjunction Risk Assessment and Maneuver Timing Optimization for Low Earth Orbit (LEO) Satellite

B.E Aerospace Engineering
Anna University Regulation 2021
SEMESTER: VIII

Academic Year: 2022-2026

DEPARTMENT OF AERONAUTICAL AND AEROSPACE ENGINEERING
KCG College of Technology,
Karapakkam, Rajiv Gandhi Salai,
Chennai – 600 097.

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BONAFIDE CERTIFICATE

This is to certify that the Report on Scalable Conjunction Risk Assessment And Maneuver Timing Optimization For Low Earth Orbit Satellite is being submitted by Satheeshkumar A(311022110039), Ignatius Nigel A(311022110020), Tharun Kumar M.S(311022110048) and Romitha R(311022110037) of the Department of Aerospace Engineering, KCG College of Technology, Chennai, as for the partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering.

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ABSTRACT

The rapid increase in the population of space debris and operational satellites in Low Earth Orbit (LEO) has significantly elevated the risk of in-orbit collisions, posing a serious threat to mission safety and long-term sustainability of the space environment. This project presents a scalable framework for conjunction risk assessment and maneuver timing optimization for LEO satellites, integrating deterministic orbital propagation, probabilistic collision modelling, and resource-efficient maneuver planning.

Orbital data of satellites and debris objects are processed using standardized Two-Line Element (TLE) sets, and future states are predicted using simplified perturbation models. A staged conjunction detection approach is implemented, beginning with coarse screening to identify potential close approaches, followed by refined Time of Closest Approach (TCA) estimation using higher-resolution analysis. Collision probability is evaluated using a three-dimensional covariance-based model in the Radial–Transverse–Normal (RTN) reference frame, incorporating uncertainties in orbital data.

The proposed system demonstrates improved computational scalability through parallel processing and optimized data handling techniques, allowing effective screening of large debris populations. Results indicate enhanced accuracy in TCA estimation and reduction in Δv expenditure through optimized maneuver timing strategies. While assumptions such as simplified covariance representation and impulsive burn models limit operational fidelity, the framework provides a robust and extensible foundation for research-grade conjunction assessment and autonomous collision mitigation systems in LEO.

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CHAPTER 1

INTRODUCTION

1.1 Background of Space Debris in LEO

Low Earth Orbit (LEO), defined as the region of space extending from approximately 160 km to 2000 km above the Earth's surface, has emerged as the most intensively utilized orbital regime in modern space operations. Its strategic importance stems from its proximity to Earth, which enables reduced communication latency, lower launch energy requirements, and enhanced spatial resolution for Earth observation missions. Consequently, LEO hosts a diverse range of satellites, including communication constellations, remote sensing platforms, scientific payloads, and defence-related systems.

However, the rapid expansion of space activities over the past few decades has led to a significant increase in the population of artificial objects in orbit. In addition to operational satellites, LEO contains a vast number of non-functional objects collectively referred to as space debris. These include defunct satellites, spent rocket stages, mission-related objects, and fragments generated from explosions and collisions.

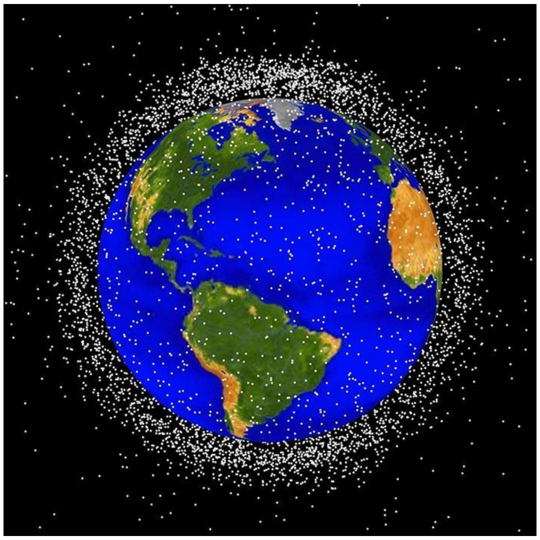

Figure 1 Growth of space debris in LEO

The debris environment in LEO is characterized by extremely high orbital velocities, typically ranging between 7 km/s and 8 km/s for individual objects. When considering relative motion between two objects in intersecting orbits, collision velocities can exceed 10–15 km/s. At such velocities, even millimeter-sized debris particles possess sufficient kinetic energy to damage or disable critical satellite components, while larger fragments can result in complete structural failure.

Several historical events have significantly contributed to the escalation of the debris population. The Fengyun-1C anti-satellite test conducted in 2007 resulted in the fragmentation of a weather satellite, generating thousands of trackable debris objects and countless smaller fragments. Similarly, the accidental Iridium 33–Cosmos 2251 collision in 2009 marked the first major collision between two active satellites, further exacerbating the debris environment.

These incidents highlight the growing concern surrounding the sustainability of space operations. One of the most critical theoretical frameworks describing this issue is the Kessler Syndrome, which predicts a cascading chain reaction of collisions, where each collision generates additional debris, thereby increasing the likelihood of further collisions. If left unmitigated, this scenario could render certain orbital regions unusable for future missions.

Given the increasing number of mega-constellations deployed by organizations such as SpaceX and other global space agencies, the density of objects in LEO is expected to rise even further. This necessitates the development of robust and scalable systems for monitoring, predicting, and mitigating collision risks in an increasingly congested orbital environment.

1.2 Problem Statement

The growing congestion in Low Earth Orbit has led to a substantial increase in the frequency of close approaches between space objects, commonly referred to as conjunction events. A conjunction event occurs when two objects in orbit approach each other within a predefined threshold distance, typically on the order of a few kilometres, depending on mission-specific safety criteria.

The primary challenge lies in accurately identifying and evaluating these conjunction events in a timely and computationally efficient manner. Current space object catalogues contain tens of thousands of tracked objects, and this number continues to grow rapidly. The computational complexity of evaluating pairwise interactions between all objects scales quadratically with the number of objects, making real-time assessment increasingly difficult.

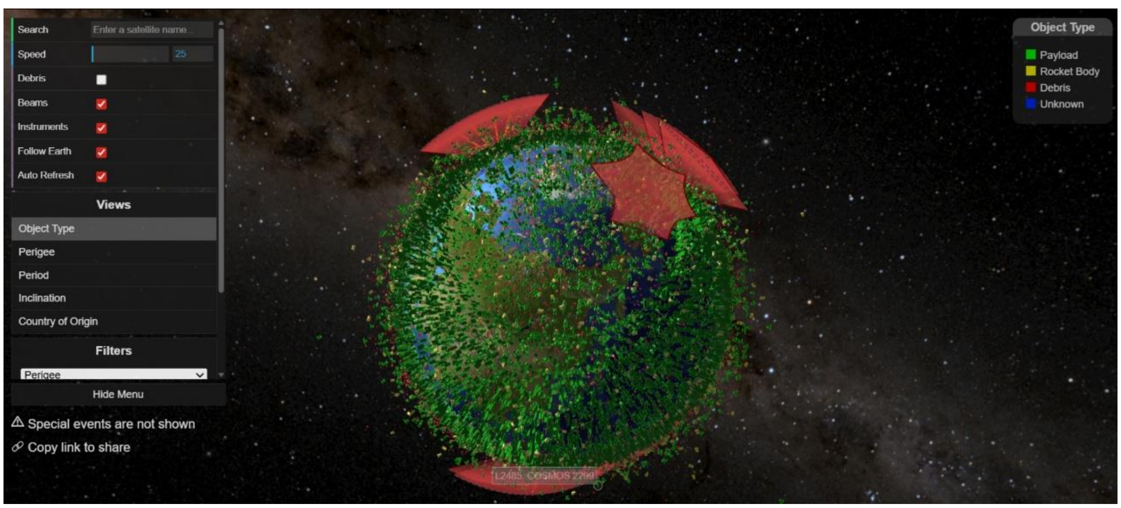

Figure 2 Overview of conjunction Risk scenario

In addition to scalability challenges, significant uncertainties exist in orbital data due to limitations in measurement accuracy, environmental perturbations, and simplifications in propagation models. These uncertainties propagate over time, leading to inaccuracies in predicted positions and velocities of space objects. As a result, determining the exact Time of Closest Approach (TCA) and the corresponding miss distance becomes a non-trivial problem.

Furthermore, even when a potential collision is detected, deciding whether to perform a collision avoidance maneuver introduces additional complexity. Maneuvers must be carefully planned to balance multiple competing objectives, including minimizing fuel consumption (Δv), maintaining mission objectives, and ensuring sufficient separation distance.

Therefore, there is a critical need for a comprehensive framework that can:

- Efficiently detect conjunction events in large-scale orbital datasets.
- Accurately estimate TCA and minimum separation distance.
- Quantify collision risk using probabilistic methods.
- Optimize maneuver strategies under operational constraints.

This project aims to address these challenges by developing a scalable and integrated approach to conjunction risk assessment and maneuver timing optimization.

1.3 Need for Conjunction Assessment

Conjunction assessment plays a central role in space situational awareness (SSA) and space traffic management (STM). It involves the continuous monitoring, prediction, and evaluation of close approaches between space objects to prevent potential collisions.

Leading space agencies such as NASA and European Space Agency operate sophisticated tracking networks and data processing systems to monitor orbital objects and issue conjunction warnings to satellite operators. These warnings are typically accompanied by key parameters such as predicted TCA, miss distance, and probability of collision.

The importance of conjunction assessment is underscored by the operational risks associated with satellite collisions. A single collision event can result in:

- Loss of valuable space assets.
- Disruption of critical services such as communication and navigation.
- Generation of additional debris, further increasing the risk environment.

Moreover, the increasing commercialization of space and the deployment of large satellite constellations have intensified the need for automated and scalable conjunction assessment systems. Manual decision-making processes are no longer sufficient to handle the volume and complexity of conjunction data.

Effective conjunction assessment enables satellite operators to make informed decisions regarding collision avoidance maneuvers. By accurately predicting high-risk events, operators can take preventive action while minimizing unnecessary maneuvers, thereby conserving limited onboard resources such as fuel.

In this context, the development of efficient, accurate, and scalable conjunction assessment techniques is essential for ensuring the long-term sustainability and safety of space operations.

1.4 Objectives of the Project

The primary objective of this project is to develop a scalable and efficient framework for conjunction risk assessment and maneuver timing optimization for satellites operating in Low Earth Orbit.

To achieve this goal, the project is structured around the following specific objectives:

- To model satellite orbits using standard orbital representations such as Two-Line Element (TLE) data.
- To implement orbit propagation using analytical models suitable for large-scale simulations.
- To design and implement a conjunction detection framework capable of handling large object catalogs.
- To refine the estimation of Time of Closest Approach (TCA) using mathematical optimization techniques.
- To compute minimum separation distances between objects with high accuracy.
- To develop a probabilistic collision risk model based on covariance analysis and Gaussian distributions.
- To design collision avoidance strategies and analyze maneuver effectiveness.
- To validate the proposed framework through simulation and performance analysis.

These objectives collectively aim to bridge the gap between theoretical models and practical implementation in real-world space operations.

1.5 Scope and Limitations

This project focuses on the development and evaluation of a conjunction risk assessment and maneuver optimization framework specifically for Low Earth Orbit scenarios.

The scope of the project includes:

- Use of publicly available orbital data such as TLE sets for representing space objects.
- Application of analytical propagation models for orbit prediction.
- Implementation of conjunction detection and refinement techniques.
- Probabilistic modeling of collision risk using simplified covariance representations.
- Design and evaluation of collision avoidance maneuvers.
- Integration of machine learning models for predictive optimization.

While the project aims to provide a comprehensive solution, certain limitations must be acknowledged. Firstly, the use of simplified propagation models may not fully capture all perturbative forces acting on satellites, such as higher-order gravitational effects, atmospheric density variations, and solar radiation pressure. These factors can introduce discrepancies between predicted and actual orbital states.

Secondly, uncertainties in TLE data and limitations in covariance estimation can affect the accuracy of collision probability calculations. The assumption of Gaussian error distributions, while common, may not always accurately represent real-world uncertainties. Additionally, the machine learning component relies on synthetically generated or limited datasets, which may impact its ability to generalize to all operational scenarios. Despite these limitations, the proposed framework provides valuable insights into scalable conjunction assessment and serves as a foundation for further research and development.

1.6 Literature Survey

The problem of space debris and collision risk in Low Earth Orbit (LEO) has been extensively studied over the past decades due to the increasing density of artificial objects in orbit. Early foundational work by Donald Kessler introduced the concept of collisional cascading, now widely known as the Kessler Syndrome, which predicts an exponential increase in debris due to successive collisions. This theory established the long-term sustainability concerns of orbital environments. Subsequent studies have highlighted the rapid growth of debris populations in LEO, particularly in Sun-synchronous orbits. Statistical analyses show that fragmentation events and satellite breakups significantly increase conjunction frequency, thereby elevating collision risk.

Conjunction assessment has therefore become a critical component of mission design and operational safety.

Modern space operations rely heavily on conjunction assessment frameworks developed by organizations such as NASA and European Space Agency. NASA's Conjunction Assessment Risk Analysis (CARA) methodology provides standardized procedures for estimating collision probability using state vectors and covariance information. These approaches model uncertainty using Gaussian distributions and reduce the collision problem to a two-dimensional encounter plane for efficient computation. Collision probability estimation has evolved from computationally intensive Monte Carlo simulations to more efficient analytical and semi-analytical models. Traditional methods assume linear relative motion and Gaussian uncertainty distributions, allowing probability computation through integral formulations. However, Monte Carlo techniques, while accurate, are often impractical for real-time applications due to high computational cost.

Recent research has focused on improving both accuracy and computational efficiency. Advanced statistical methods and reformulations of collision probability equations have enabled faster computation while maintaining precision, making them suitable for real-time conjunction assessment systems. Additionally, studies emphasize the importance of covariance accuracy, as uncertainty in position and velocity significantly affects collision probability estimation. Furthermore, alternative approaches have been proposed to address limitations in traditional probability-based metrics. Statistical inference methods based on miss distance estimation provide improved interpretability in certain scenarios, especially when uncertainty is high. These methods offer complementary insights to conventional probability-based risk assessment.

From an operational perspective, collision avoidance strategies involve maneuver planning based on predicted conjunction events. Satellite operators routinely perform orbit adjustments to reduce collision risk, especially as conjunction frequency increases due to the proliferation of satellite constellations. Studies indicate that without active avoidance maneuvers, collision events in LEO could occur within very short time intervals, highlighting the urgency of robust detection and mitigation systems

Recent literature also emphasizes the need for integrated approaches combining physics-based models with data-driven techniques. Machine learning methods are being explored to accelerate prediction and optimization tasks, particularly in large-scale environments involving thousands of objects. These hybrid approaches aim to enhance real-time decision-making while maintaining acceptable accuracy levels.

Overall, the literature indicates that while significant progress has been made in conjunction detection, collision probability estimation, and avoidance strategies, challenges remain in handling uncertainty, scalability, and real-time computation.

CHAPTER 2

ORBITAL MECHANICS AND CONJUNCTION THEORY

2.1 Fundamentals of Orbital Motion

To understand conjunction risk, we first need to understand how objects move in space. Unlike motion on Earth, orbital motion is governed almost entirely by gravity and inertia, creating a delicate balance that keeps satellites in continuous free fall around the Earth.

At the heart of orbital mechanics lies **Newton's Law of Universal Gravitation**, which states that every object attracts every other object with a force proportional to their masses and inversely proportional to the square of the distance between them. In the context of satellites, the Earth provides the gravitational pull, while the satellite's velocity provides the forward motion.

Table 1 Orbital Parameters used in Simulation

Parameter	Symbol	Value	Unit
Semi-major axis	a	6878	km
Eccentricity	e	0.001	—
Inclination	i	98.5	deg
RAAN	Ω	120	deg
Argument of Perigee	ω	85	deg
True Anomaly	ν	45	deg

This balance results in different types of orbits:

- Circular orbits (constant altitude).
- Elliptical orbits (varying altitude).
- Inclined orbits (tilted relative to Earth's equator).

Each orbit is defined using **Keplerian elements**, derived from **Kepler's Laws of Planetary Motion**, which include:

- Semi-major axis (size of orbit).
- Eccentricity (shape of orbit).
- Inclination (tilt).
- Right Ascension of Ascending Node (orientation).
- Argument of Perigee.
- True anomaly (position of satellite).

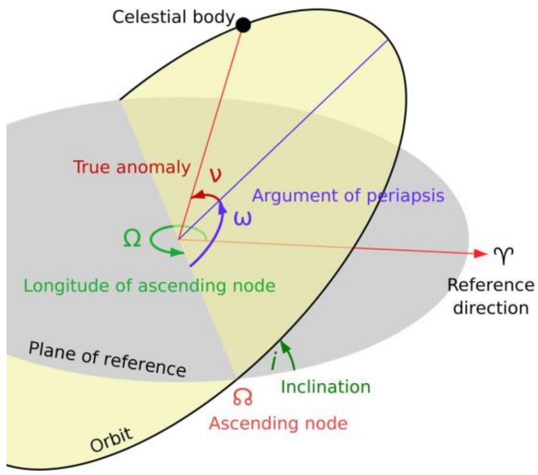

Figure 3 Orbital Elements Representation

In simple terms, orbital motion is predictable but sensitive. Even small disturbances like atmospheric drag or gravitational effects from the Moon can slowly alter a satellite's path over time. This is exactly why predicting future positions becomes challenging and directly impacts conjunction assessment.

2.2 Two-Line Element (TLE) Representation

In real-world applications, we don't manually calculate orbital elements every time. Instead, satellites are tracked using a standardized format known as the **Two-Line Element (TLE)** set.

A TLE is essentially a compact data representation that encodes all the necessary orbital parameters of a space object into two lines of text. It is widely used by organizations such as NORAD and distributed through catalogs maintained by CelesTrak.

A typical TLE contains:

- Satellite identification number.
- Epoch time (reference time for orbit).
- Inclination.
- Right Ascension of Ascending Node.
- Eccentricity.
- Argument of perigee.
- Mean anomaly.
- Mean motion (revolutions per day).

What makes TLEs powerful is their simplicity and accessibility. However, they come with limitations:

- They are approximations, not exact states.
- Accuracy degrades over time.
- They depend on specific propagation models.

Table 2 TLE Sample Data

Line	Data
Line 1	1 25544U 98067A 21275.51843750 .00002182 00000-0 49618-4 0 9993
Line 2	2 25544 51.6442 21.4473 0004201 35.1234 45.6789 15.489876543210

2.3 SGP4 Propagation Model

To move from a static TLE to a dynamic orbit prediction, we use propagation models. The most widely used model for TLE-based propagation is the **Simplified General Perturbations 4 (SGP4)** model.

SGP4 is not just a basic physics equation—it is a carefully engineered model that accounts for key perturbations affecting satellites in LEO. These include:

- Earth’s non-spherical shape (oblateness).
- Atmospheric drag.
- Gravitational perturbations.

The model is used extensively by organizations such as NASA and the United States Space Force for tracking space objects.

Conceptually, SGP4 works as follows:

1. It takes a TLE as input.
2. Applies perturbation corrections.
3. Outputs position and velocity at a given time.

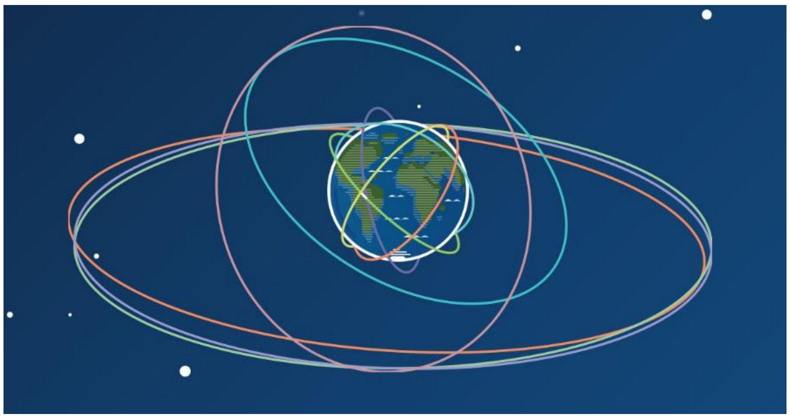

Figure 4 SGP4 Propagation model

However, it is important to understand its limitations:

- It is optimized for speed, not perfect accuracy.
- Best suited for short-term predictions (a few days).
- Errors increase over longer time spans.

2.4 Relative Motion Between Objects

Conjunction analysis is not about a single satellite it's about how two objects move relative to each other.

Even if two satellites are in stable orbits, their paths may intersect or come dangerously close due to differences in:

- Orbital altitude.
- Inclination
- Phase (position along orbit)

To analyze this, we shift from an Earth-centered view to a **relative frame of reference**, where one object is considered stationary, and the motion of the other is observed relative to it.

This relative motion can be:

- Linear (for short durations)
- Curved (over longer durations)

One commonly used approximation is based on linearized relative motion equations, which simplify analysis near conjunction events.

A key insight here is:

Two objects do not need to collide directly to be dangerous proximity **itself is the risk**.

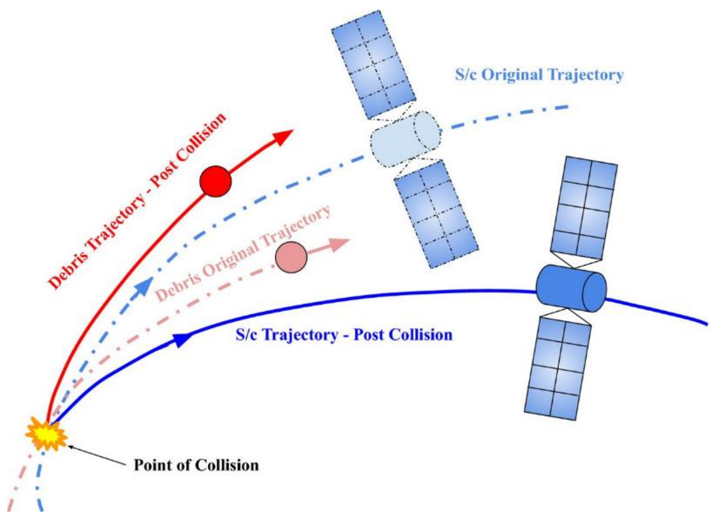

Figure 5 Relative motion Between Satellite and Debris

2.5 Time of Closest Approach (TCA)

The **Time of Closest Approach (TCA)** is one of the most critical parameters in conjunction analysis. It represents the exact moment when two objects are at their minimum separation distance.

- Two satellites are moving along their own paths.
- At some point, their distance becomes smallest.
- That instant is the TCA.

Mathematically, TCA is found by minimizing the relative distance between two objects as a function of time.

Why is TCA important?

- It defines the peak collision risk moment.
- It is used to compute collision probability.
- It determines when a maneuver should occur.

However, calculating TCA is not always straightforward:

- Orbital paths are nonlinear.
- Uncertainties affect predictions.
- High precision is required.

This is why refinement techniques (like polynomial fitting) are used after initial detection.

2.6 RTN Coordinate Frame

To better understand and analyze relative motion, we use a specialized coordinate system called the **RTN frame** (Radial, Transverse, Normal).

This frame is centered on the satellite and moves along with it:

- **Radial (R):** Points outward from Earth to the satellite.
- **Transverse (T):** Tangential to the orbit (direction of motion).
- **Normal (N):** Perpendicular to the orbital plane.

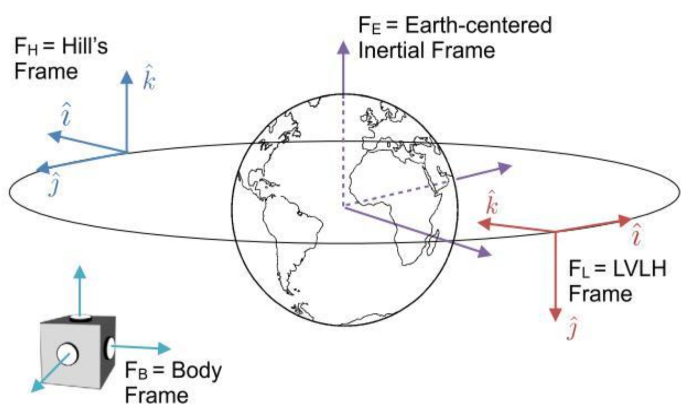

Figure 6 RTN Coordinate System

Because it transforms a complex 3D motion problem into a more intuitive structure:

- Radial → altitude differences.
- Transverse → along-track differences.
- Normal → plane separation.

In collision analysis, uncertainties are also represented in this frame, making it easier to model how errors affect position.

RTN is widely used in:

- Conjunction analysis.
- Maneuver design.
- Covariance transformation.

2.7 Sources of Orbital Uncertainty

No matter how advanced our models and tracking systems are, **uncertainty is an inherent part of space operations**. Unlike controlled environments on Earth, space is influenced by multiple dynamic and partially unpredictable factors. These uncertainties directly affect our ability to accurately predict satellite positions and, consequently, impact conjunction assessment and collision risk estimation.

Even a small positional error can grow significantly over time, especially in Low Earth Orbit (LEO), where objects move at extremely high velocities. Therefore, understanding the sources of uncertainty is essential for building reliable prediction and risk assessment systems.

1. Measurement Errors

All orbital predictions begin with measurements obtained from tracking systems such as radar and optical sensors. However, these systems are not perfect and introduce errors during data acquisition. Radar systems, for example, measure range, velocity, and direction, but are affected by noise, signal interference, and resolution limitations. Optical sensors rely on light detection, which can be influenced by atmospheric conditions, lighting, and observation angles. These measurement errors translate into inaccuracies in estimating the satellite's position and velocity vectors. Since orbital propagation depends heavily on initial conditions, even a small error at the starting point can lead to significant deviations over time. Additionally, tracking is not continuous for all objects. Some debris may only be observed intermittently, leading to gaps in data and further uncertainty in state estimation.

2. Atmospheric Drag

In Low Earth Orbit, satellites are still affected by the presence of a very thin layer of Earth's atmosphere. Although extremely tenuous, this residual atmosphere exerts a drag force that gradually reduces the satellite's velocity and altitude.

The complexity arises because atmospheric density is **not constant**. It varies due to:

- Solar activity (solar flares, geomagnetic storms).
- Time of day (day-night heating differences).

- Seasonal and geographic variations.

During periods of high solar activity, the atmosphere expands, increasing drag significantly. This causes satellites to lose altitude faster than expected. Since atmospheric models cannot perfectly predict these variations, drag becomes one of the **largest sources of uncertainty in LEO orbit prediction**. Over time, this leads to increasing errors in position and timing of orbital passes.

3. Gravitational Perturbations

While basic orbital motion assumes Earth as a perfect sphere, Earth has an irregular shape with uneven mass distribution. This is due to factors such as:

- Mountain ranges.
- Ocean trenches.
- Variations in Earth's structure.

This non-uniformity leads to what are called **gravitational perturbations**. The most significant among these is the Earth's oblateness (flattening at the poles), often represented by the J2 perturbation.

These perturbations cause gradual changes in orbital parameters, such as:

- Precession of the orbital plane.
- Drift in the argument of perigee.
- Changes in inclination over time.

In addition to Earth's gravity, external bodies like the Moon and the Sun also exert gravitational forces, further altering satellite trajectories.

Although models attempt to account for these effects, they cannot capture every detail perfectly, leading to accumulated uncertainty over time.

4. Solar Radiation Pressure

Solar radiation pressure is caused by photons emitted from the Sun exerting force on a satellite's surface. Although the force is extremely small, it becomes significant over long durations, especially for satellites with:

- Large surface areas (e.g, solar panels).
- Low mass (higher area-to-mass ratio).

This force can cause gradual changes in orbit, including:

- Drift in position.
- Changes in orbital energy.
- Small deviations in trajectory.

The effect is also dependent on:

- Satellite orientation (attitude).
- Reflectivity of surfaces.
- Shadowing effects (eclipses).

Since these factors vary dynamically and are difficult to model precisely, solar radiation pressure contributes to long-term prediction errors.

5. Model Limitations

Orbital propagation models such as SGP4 are designed to balance **accuracy and computational efficiency**. While they incorporate major perturbative effects, they rely on simplifying assumptions.

- Atmospheric drag is modeled using average conditions.
- Higher-order gravitational effects may be approximated.
- Complex interactions are simplified.

These simplifications make the model fast and practical but introduce approximation errors. Over short durations, these errors are minimal, but over longer periods, they accumulate and reduce prediction accuracy.

Thus, even with a perfect initial state, the model itself introduces uncertainty.

6. TLE Inaccuracies

Two-Line Element (TLE) data is widely used for orbit representation due to its simplicity and availability. However, TLEs are not exact measurements they are estimates derived from observational data.

Key limitations of TLEs include:

- Limited precision (rounded parameters).
- Dependence on observation quality.
- Updates at discrete intervals.

As time progresses, the accuracy of a TLE degrades because it does not account for all dynamic changes affecting the satellite.

This means:

- Older TLE → higher prediction error.
- Frequent updates → better accuracy.

In conjunction analysis, using outdated TLE data can significantly affect TCA estimation and collision probability calculations.

Why Uncertainty Matters

Uncertainty fundamentally changes how we approach collision analysis. Instead of representing a satellite as a single point in space, we represent it as a **region of possible positions**. This region is typically modeled as an ellipsoid derived from the covariance matrix.

Implications of Uncertainty

- Collision is not deterministic — it cannot be predicted with absolute certainty.
- We compute a **probability of collision**, not a definite outcome.
- Decision-making becomes **risk-based rather than absolute.**

Even if two objects do not appear to collide based on nominal trajectories, their uncertainty regions may overlap, leading to a non-zero collision probability.

CHAPTER 3

CONJUNCTION DETECTION AND TCA REFINEMENT

3.1 Conjunction Detection Framework

Imagine monitoring thousands of satellites and debris fragments, all moving at extremely high velocities often exceeding 7–8 km/s in Low Earth Orbit. Each object follows its own orbital path, influenced by gravitational forces, atmospheric drag, and other perturbations. In such a dynamic and densely populated environment, even a slight overlap in trajectories can result in a potentially hazardous situation.

The objective of conjunction detection can be expressed simply: to identify which pairs of space objects may come dangerously close to each other and determine when this close approach will occur. While this idea appears straightforward, its practical implementation is highly complex due to the scale of the problem and the continuous nature of orbital motion. To address this complexity, conjunction detection is performed using a structured, multi-stage framework that progressively refines the analysis while maintaining computational efficiency.

The first stage is orbit propagation. In this step, the future positions and velocities of all objects are predicted over a given time window. Using propagation models such as SGP4, each object's orbit is advanced from its initial state, typically derived from Two-Line Element data. This produces time-dependent position and velocity vectors for every object in the system. This step is fundamental because conjunction events are future occurrences, and accurate prediction of motion is essential for identifying them.

Once the orbital states are propagated, the next step is coarse screening. In a system containing thousands of objects, it is not practical to evaluate every possible pair in detail. Therefore, coarse screening is used to quickly eliminate object pairs that are clearly too far apart to pose any risk. This is typically achieved through spatial filtering, approximate distance checks, or the use of simplified bounding regions around orbits. The goal of this stage is not precision, but efficiency removing many irrelevant pairs early in the process.

After coarse screening, a reduced set of candidate pairs remains. These pairs are then analysed in the fine screening or detection stage. Here, the relative motion between objects is examined more closely by computing the distance between them over time. If the distance falls below a predefined threshold, the pair is flagged as a potential conjunction. This stage provides a more accurate identification of close approaches and defines a time window within which the closest interaction is likely to occur.

However, at this stage, the exact moment of closest approach is still not precisely known. To determine this, the framework proceeds to TCA refinement. Instead of relying solely on discrete time samples, mathematical techniques such as polynomial approximation are used to model the distance between objects as a continuous function of time. By analyzing this function, the exact time at which the distance is minimized can be determined with greater accuracy. This refinement is crucial because even small timing errors can lead to significant position errors in high-speed orbital motion.

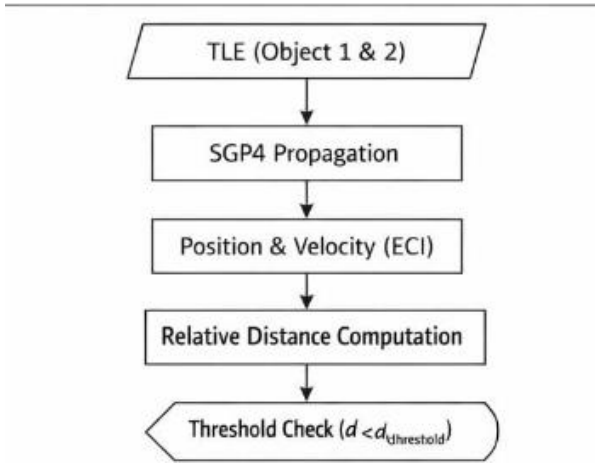

Figure 7 Conjunction Detection Workflow

Once the Time of Closest Approach and minimum separation distance are obtained, the final step in the framework is risk evaluation. At this point, the severity of the conjunction is assessed by determining whether the objects are likely to collide and whether a collision avoidance maneuver is necessary. This step connects directly to the probabilistic modeling discussed in the next chapter.

A major challenge in conjunction detection is scalability. If there are N objects in orbit, the total number of possible object pairs is given by N(N−1)/2. This relationship grows rapidly as the number of objects increases. For example, a system with 1,000 objects results in approximately 500,000 possible pairs, while 10,000 objects result in nearly 50 million pairs. Evaluating each of these pairs with high precision would require enormous computational resources.

Because of this, efficient filtering is essential. The framework relies on multi-stage screening and approximation techniques to reduce the number of pairs that require detailed analysis. By eliminating obviously safe pairs early, the system ensures that computational effort is focused only on potentially hazardous interactions.

The key insight is that effective conjunction detection is not about analyzing every possible interaction in detail, but about intelligently narrowing down the problem to a manageable set of high-risk candidates.

In conclusion, the conjunction detection framework provides a scalable and efficient approach to identifying potential close approaches in a crowded orbital environment. By combining orbit propagation, coarse and fine screening, and precise refinement techniques, the framework forms a strong foundation for accurate collision risk assessment and subsequent maneuver planning.

3.2 Coarse Screening Method

Before performing computationally intensive calculations for conjunction detection, it is essential to reduce the number of object pairs that need to be analyzed in detail. This initial filtering stage is known as coarse screening.

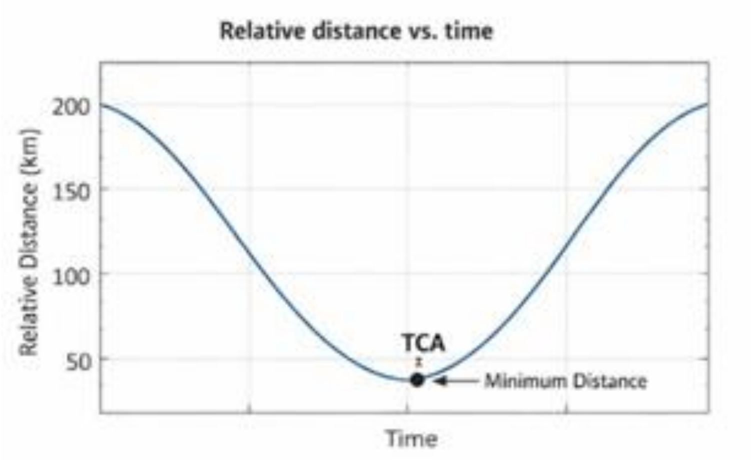

Figure 8 Relative Distance vs time curve

The core idea behind coarse screening is simple: instead of evaluating every possible interaction, we first eliminate pairs that are clearly too far apart to pose any collision risk. In practical terms, this step answers a fundamental question: *"Are these two objects even close enough to worry about?"* By applying fast and approximate filtering techniques, coarse screening significantly reduces the computational burden of the overall system.

Table 3 Screening Threshold Parameters

Parameter	Value	Unit
Distance Threshold	50	km
Time Step (Coarse)	60	sec
Time Step (Refined)	1	sec
Screening Radius	100	km

Spatial Filtering

Spatial filtering groups objects based on their orbital characteristics such as altitude, inclination, and orbital plane. Objects that belong to entirely different orbital regions are highly unlikely to interact.

For example, a satellite in a low-inclination equatorial orbit will rarely intersect with an object in a polar orbit. Similarly, large differences in altitude reduce the likelihood of close approaches. By organizing objects into spatial zones, only those within similar regions are compared, which minimizes unnecessary computations while still preserving potentially risky pairs.

Bounding Volume Method

In this method, each object's trajectory is approximated using a simplified geometric shape such as a sphere, box, or ellipsoid. These shapes represent the region of space that the object may occupy over a given time interval.

Instead of calculating exact distances between objects, we first check whether their bounding volumes overlap. If these volumes do not intersect, it guarantees that the objects themselves cannot come close to each other. This approach is highly efficient because geometric overlap checks are much faster than detailed orbital calculations.

Time Window Filtering

Conjunction events depend not only on spatial proximity but also on timing. Two objects may pass through the same region of space but at completely different times, meaning there is no risk of collision.

Time window filtering ensures that only objects that are close in both space and time are considered. If their time intervals do not overlap, the pair is discarded. This reduces unnecessary computations and prevents false identification of conjunction risks.

Why Coarse Screening Matters

The importance of coarse screening becomes evident when considering the scale of conjunction detection. As the number of objects increases, the number of possible pairs grows rapidly, making direct evaluation impractical.

Coarse screening plays a critical role by reducing computational load significantly, filtering out a large percentage of irrelevant pairs, enabling near real-time processing, and making large-scale conjunction analysis feasible.

Key Insight

Coarse screening is designed to be conservative. Its purpose is not to detect conjunctions precisely, but to ensure that no potentially dangerous pairs are missed. Some safe pairs may still pass through to later stages, which is acceptable because further refinement will handle them.

Conclusion

Coarse screening forms the backbone of an efficient conjunction detection system. By combining spatial grouping, geometric approximations, and time-based filtering, it effectively reduces problem complexity while preserving all relevant candidate interactions. Without this step, conjunction analysis for large object catalogs would be computationally impractical.

3.3 Distance-Based Detection

After coarse filtering, we move to **distance-based detection**, where actual relative distances between objects are evaluated over time.

Core Idea:

We track how the distance between two objects changes as they move along their orbits.

Let:

- $\mathbf{r}_1(t)$= position of object 1
- $\mathbf{r}_2(t)$= position of object 2Then, the relative distance is:

Then, the relative distance is:

$$d(t) = \| \mathbf{r}_1(t) - \mathbf{r}_2(t) \|$$

conjunction is detected if:

d(t) < d threshold

where (d{threshold}) is typically in the range of **1–5 km**.

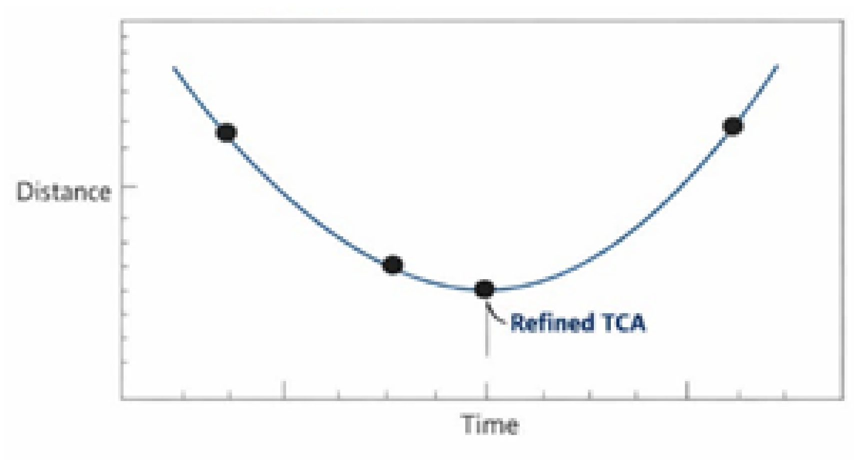

Figure 9 Quadratic TCA Refinement

Practical Insight

Instead of checking continuously, we:

- Sample positions at discrete time steps.
- Track how distance evolves.
- Identify potential "close approach regions".

This gives a **rough estimate** of when the objects might come closest but not the exact TCA yet.

3.4 Polynomial-Based TCA Refinement

After identifying a potential conjunction using distance-based detection, the next critical step is to determine the **exact Time of Closest Approach (TCA)** with high precision. The initial detection stage typically provides only an approximate time window where the objects come close. However, for accurate collision risk assessment, a more refined estimate is required.

Table 4 TCA Detection Accuracy Comparison

Method	Time Error (sec)	Distance Error (m)
Coarse Sampling	10	500
Linear Approximation	3	120
Quadratic Refinement	0.5	10

Polynomial-based TCA refinement is used to achieve this higher level of precision by modeling the relative distance between two objects as a continuous function of time.

Concept of Refinement

During the detection stage, the distance between two objects is evaluated at discrete time intervals. While this helps identify a close approach, it does not guarantee that the exact minimum distance point is captured, since the true minimum may lie between sampled points.

To overcome this limitation, we focus on a small-time interval around the suspected closest approach and approximate the distance variation using a polynomial function. This allows us to estimate the minimum point more accurately without requiring extremely fine time sampling.

Quadratic Approximation

The most used approach is quadratic approximation. In this method, the distance function is approximated as a second-degree polynomial:

$$d(t) \approx at2 + bt + c$$

This approximation is valid within a short time window around the closest approach, where the motion can be considered smooth and continuous.

The minimum of this quadratic function corresponds to the Time of Closest Approach. It can be obtained analytically by finding the vertex of the parabola:

$$t_{TCA} = -\frac{b}{2a}$$

This provides a precise estimate of the time at which the distance between the two objects is minimized.

Refinement Process

The refinement process typically involves selecting three or more data points around the suspected closest approach. These points are used to fit a quadratic curve that represents the distance variation with time.

Once the polynomial is fitted, the minimum point is calculated analytically. The corresponding time gives the refined TCA, and substituting this time back into the relative position equation provides the minimum separation distance.

This approach significantly improves accuracy compared to discrete sampling methods.

Advantages of Polynomial Refinement

Polynomial-based refinement offers several advantages. It is computationally efficient because it avoids excessive sampling while still providing high precision. It is also mathematically simple and can be implemented easily in real-time systems.

Additionally, it captures the local curvature of the distance function, which is essential for identifying the true minimum point.

Limitations

Despite its advantages, polynomial refinement has certain limitations. The quadratic approximation is only valid within a small-time window and may not accurately represent highly nonlinear motion over longer durations.

The accuracy of the method also depends on the quality of the initial detection. If the selected time window does not properly contain the true minimum, the refinement may produce inaccurate results.

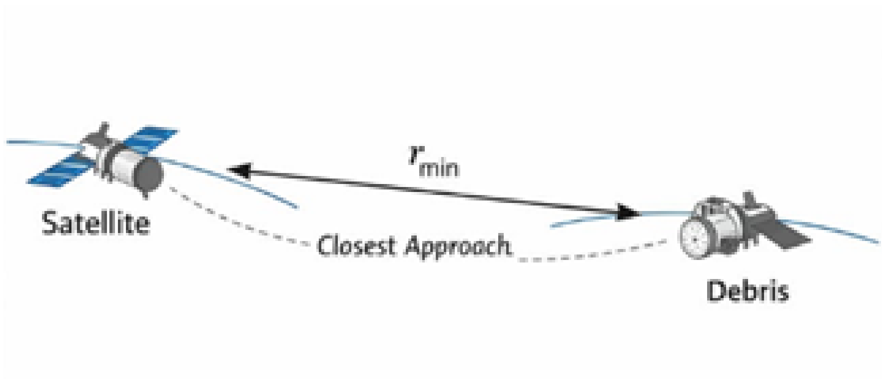

Figure 10 Minimum Distance Identification

Furthermore, uncertainties in orbital data and propagation models can affect the precision of the refined TCA.

Importance in Conjunction Analysis

Accurate estimation of TCA is critical for subsequent steps in conjunction assessment. Collision probability calculations and maneuver planning both rely heavily on precise timing and minimum distance values.

Even small errors in TCA estimation can lead to significant differences in predicted separation due to the high relative velocities involved. Therefore, polynomial-based refinement plays a vital role in ensuring reliable conjunction analysis.

Conclusion

Polynomial-based TCA refinement provides an efficient and accurate method for determining the exact time and distance of closest approach between two objects. By approximating the distance function using a quadratic model, it enables precise identification of the minimum point while maintaining computational efficiency. This makes it an essential component of modern conjunction detection frameworks.

3.5 Minimum Distance Estimation

Once the Time of Closest Approach (TCA) is accurately determined, the next step is to compute the **minimum separation distance** between the two objects. This value represents the shortest distance between their trajectories during the conjunction event and serves as a key indicator of potential collision risk.

The minimum distance is calculated by evaluating the relative position of the two objects at the refined TCA. Since both objects are moving at high velocities, even a small timing error can significantly affect the estimated separation. Therefore, this calculation must be performed with high precision.

This parameter is critical because it directly indicates how close the objects come to each other and forms the basis for further risk assessment. It is also a primary input for collision probability models, where the likelihood of collision is evaluated based on both separation distance and uncertainty.

Important Distinction

It is important to understand that a small separation distance does not necessarily imply a collision. Two objects may pass very close to each other without colliding, depending on their relative geometry and uncertainties.

However, as the separation distance decreases, the risk of collision increases significantly. Therefore, minimum distance is treated as a key risk indicator rather than a definitive measure of collision.

Practical Considerations

The accuracy of minimum distance estimation depends on several factors. One of the most important is the quality of the orbital propagation model used. Any errors in predicted position or velocity will directly affect the computed separation.

The estimation is also highly sensitive to uncertainties in the orbital data. Even small errors in initial conditions can lead to noticeable deviations at the time of closest approach.

To improve interpretability, minimum distance is often analyzed in the RTN (Radial, Transverse, Normal) coordinate frame. This allows the separation to be broken down into meaningful components, such as along-track and cross-track distances, providing better insight into the nature of the conjunction.

Example Insight

In practical conjunction analysis, different ranges of separation distance correspond to different levels of concern. A separation of around 10 km is generally considered safe and does not require further action. When the distance reduces to around 1 km, the situation requires closer monitoring and possibly further analysis. If the separation falls below 100 meters, it is typically classified as a high-risk conjunction, and collision avoidance maneuvers are strongly considered.

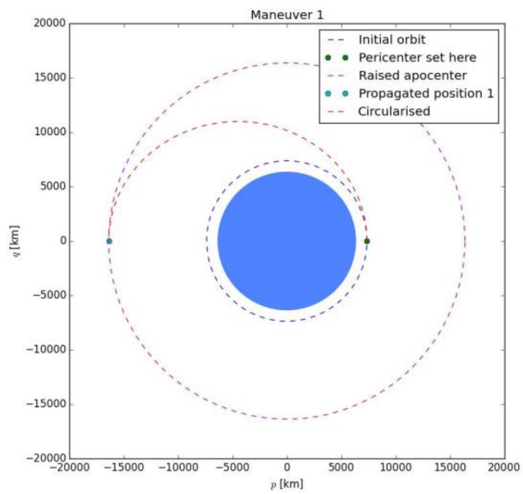

Figure 11 example maneuver visualization

Conclusion

Minimum distance estimation plays a crucial role in bridging detection and risk assessment. By accurately quantifying how close two objects come to each other, it provides a clear basis for evaluating collision risk and deciding whether mitigation actions are required.

3.6 Computational Efficiency

The entire conjunction detection process, from orbit propagation to TCA refinement and minimum distance estimation, must be highly optimized to handle real-world scenarios. Modern space surveillance systems track tens of thousands of objects, resulting in millions of potential object pairs that need to be analyzed.

Without efficient algorithms and optimization techniques, conjunction detection would become computationally infeasible.

Challenges

One of the primary challenges is the massive number of object pairs that must be evaluated. As the number of tracked objects increases, the number of possible interactions grows rapidly, leading to a significant computational burden.

Another challenge is the need for continuous orbit propagation over time. Since conjunction events can occur at any point within a prediction window, the system must evaluate object positions across multiple time steps.

Additionally, high precision requirements further increase computational complexity. Accurate estimation of TCA and minimum distance requires refined calculations, which can be expensive when applied to large datasets.

Optimization Strategies

To address these challenges, several optimization strategies are employed.

Multi-stage filtering is one of the most effective techniques. By combining coarse screening and fine detection methods, the system reduces the number of object pairs that require detailed analysis. This ensures that computational effort is focused only on potentially risky interactions.

Time step optimization is another important strategy. Instead of using a uniform time resolution throughout the simulation, adaptive time steps are used. Larger time steps are applied during initial screening to quickly identify potential conjunctions, while smaller time steps are used near suspected close approaches to improve accuracy.

Parallel processing is widely used to improve performance. Since the evaluation of different object pairs is independent, computations can be distributed across multiple processors or cores. This significantly reduces overall processing time and enables near real-time analysis.

Efficient data structures also play a key role in improving performance. Techniques such as spatial indexing, grid-based partitioning, and tree structures help organize objects in a way that reduces search complexity. This allows the system to quickly identify nearby objects without scanning the entire dataset.

Real World Relevance

In operational environments, organizations such as NASA and European Space Agency rely on highly optimized conjunction detection pipelines. These systems can process massive datasets in near real-time, ensuring timely detection of potential collision risks.

Conclusion

Computational efficiency is a fundamental requirement for scalable conjunction detection. By combining filtering techniques, adaptive algorithms, parallel processing, and efficient data handling, the system can manage large-scale space object catalogs while maintaining accuracy. These optimizations ensure that conjunction analysis remains practical and reliable in increasingly congested orbital environments.

CHAPTER 4

COLLISION PROBABILITY MODELING

4.1 Probabilistic Risk Assessment

After identifying a conjunction and estimating the Time of Closest Approach (TCA), the next crucial question that arises is whether the objects will collide. While this may initially appear to be a simple yes-or-no question, real-world space operations reveal that the situation is far more complex due to the presence of uncertainty in orbital prediction.

In an ideal scenario, if the exact positions and velocities of both objects were known without any error, it would be possible to determine deterministically whether a collision would occur. However, in practice, this level of precision is not achievable. As a result, collision assessment must be approached differently.

Deterministic vs Probabilistic Approach

Traditionally, problems in physics are solved deterministically, where a known set of inputs produces a definite outcome. However, in space operations, the inputs themselves are uncertain. Measurements are not exact, models are approximations, and environmental influences are dynamic.

Because of this, collision assessment cannot rely on a deterministic approach. Instead of asking “Will the objects collide?”, the problem is reframed to evaluate the likelihood of collision. The question becomes “What is the probability that the objects will collide?” This probabilistic approach provides a more realistic and practical way of handling uncertainty.

Table 5 Covariance Parameters

Parameter	Value	Unit
σ_R (Radial)	50	m
σ_T (Along-track)	100	m
σ_N (Cross-track)	75	m

Representation of Uncertainty

In this framework, each object is no longer treated as a single point in space. Instead, it is represented as a region of possible positions around its predicted trajectory. This region accounts for uncertainties in measurement and prediction.

The position of the object is therefore modeled as a distribution rather than a fixed value. When two such uncertain objects are considered, the potential for collision depends on how these regions overlap at the time of closest approach.

Sources of Uncertainty

The need for probabilistic modeling arises from multiple sources of uncertainty present in space operations.

Measurement errors are one of the primary contributors. Tracking systems such as radar and optical sensors provide estimates of position and velocity, but these measurements are affected by noise, resolution limits, and observational constraints.

Propagation models also introduce uncertainty. Models used to predict orbital motion simplify complex physical interactions and cannot perfectly capture all perturbations. Over time, these small inaccuracies accumulate, leading to deviations in predicted positions.

External forces further complicate prediction. Atmospheric drag varies with solar activity and atmospheric conditions. Solar radiation pressure depends on the satellite's physical properties and orientation. Gravitational perturbations arise from irregularities in Earth's mass distribution as well as the influence of the Moon and Sun.

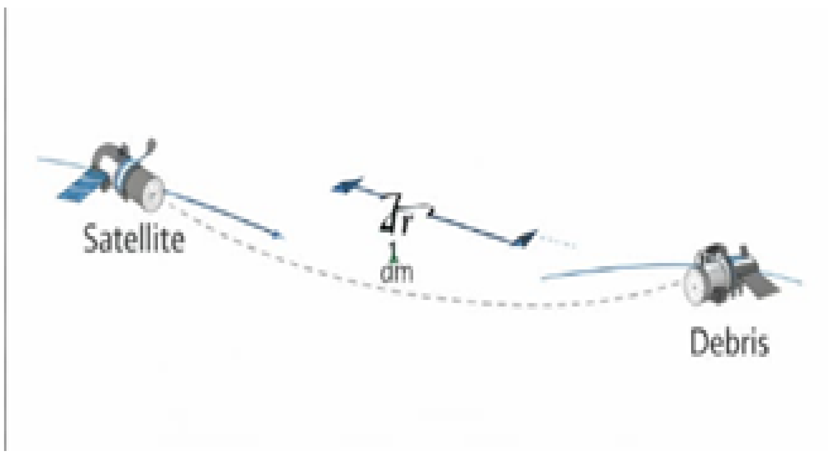

Figure 12 Covarience Geometry at TCA

Risk-Based Decision Making

The probabilistic approach enables operators to quantify collision risk in numerical terms. This allows decisions to be made based on predefined risk thresholds rather than assumptions.

For example, a very low probability of collision may not require any action, while a higher probability may trigger a collision avoidance maneuver. This ensures that maneuvers are performed only when necessary, optimizing fuel usage and mission efficiency.

This approach is widely adopted by organizations such as NASA and the European Space Agency for operational decision-making.

Importance in Space Operations

Probabilistic risk assessment plays a critical role in modern space situational awareness. With thousands of objects in orbit, it is not feasible to treat every close approach as a guaranteed collision.

By quantifying uncertainty and expressing risk in probabilistic terms, operators can prioritize conjunction events and allocate resources effectively. This approach ensures both safety and efficiency in an increasingly congested orbital environment.

Conclusion

In conclusion, probabilistic risk assessment provides a realistic and practical framework for evaluating collision risk. By accounting for uncertainties in measurement, modeling, and environmental conditions, it transforms conjunction analysis from a binary prediction problem into a risk-based decision-making process. This approach is essential for maintaining the safety and sustainability of space operations.

4.2 Covariance Representation

To mathematically represent uncertainty in the position and velocity of space objects, a covariance matrix is used. This matrix provides a compact and precise way to describe how uncertain an object's state is and how these uncertainties vary in different directions.

Intuition Behind Covariance

In real-world space operations, we cannot say that a satellite is located at an exact point in space. Instead, we describe its position in terms of likelihood.

"The satellite is most likely here, but it could be slightly displaced in different directions. "This means the satellite's position is not a fixed point, but a region of possible locations around a predicted value. Covariance helps quantify this uncertainty in a structured mathematical form.

What is Covariance?

Covariance is a statistical measure that describes how variables vary with respect to each other. In the context of orbital mechanics, it represents how uncertainties in position and velocity are distributed across different axes.

A covariance matrix captures two important aspects:

- Variance along each axis, which indicates the magnitude of uncertainty in that direction.
- Correlation between axes, which describes how uncertainties in one direction are related to uncertainties in another.

In three-dimensional space, the covariance matrix is typically represented as a 3×3 matrix for position, or a 6×6 matrix when both position and velocity are considered.

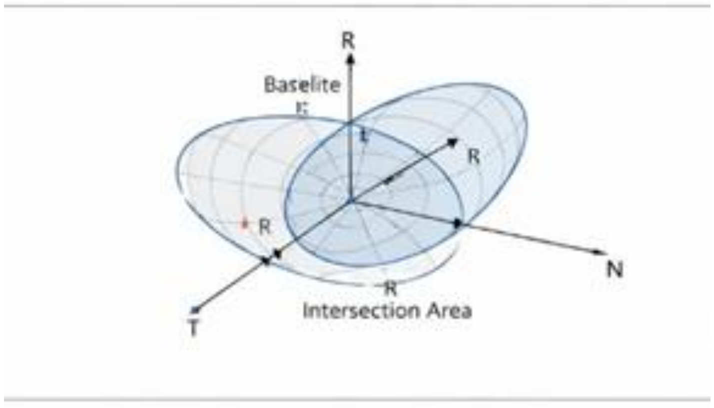

Figure 13 Covariance Ellipsoid Representation

In 3D space, this is represented as:

P =

$[\sigma x2 \quad \cdots \quad \cdots$

$\cdots \quad \sigma y2 \quad \cdots$

$\cdots \quad \cdots \quad \sigma z2]$

Visualization

This uncertainty forms a **3D ellipsoid** around the predicted position:

- Larger ellipsoid → higher uncertainty
- Smaller ellipsoid → more precise prediction

Table 6 Collision probability Results

Separation Distance (m)	Probability of Collision
1000	1e-6
500	1e-5
100	1e-3
50	1e-2

Key Insight

Each object has its own uncertainty.
For conjunction analysis, we combine them into a **relative covariance**.

4.3 RTN Frame Transformation

Working in Earth-centered inertial (ECI) or Earth-centered Earth-fixed (ECEF) coordinate systems is useful for general orbit representation. However, for conjunction analysis and collision probability modeling, these global coordinate frames are not always intuitive. This is because they do not directly reflect how objects move relative to each other along their orbital paths.

To simplify the analysis of relative motion and uncertainty, the coordinate system is transformed into a local reference frame known as the RTN frame (Radial, Transverse, Normal). This frame is centered on the satellite and moves along with it, making it more suitable for understanding conjunction geometry.

Why RTN Frame is Used

The primary reason for using the RTN frame is that it aligns naturally with the satellite's motion. Instead of describing motion in a fixed global coordinate system, the RTN frame provides a moving coordinate system that is directly tied to the satellite's trajectory.

This alignment allows us to interpret relative motion and uncertainty in a more meaningful way. For example, it becomes easier to distinguish whether two objects are separated along their orbital path, across the orbit, or radially away from Earth.

Definition of RTN Axes

The RTN frame consists of three mutually perpendicular axes:

The Radial (R) axis points from the center of the Earth directly toward the satellite. It represents the direction of altitude and captures variations in radial distance. The Transverse (T) axis lies in the direction of motion of the satellite, tangent to its orbit. It represents along-track motion and is strongly influenced by velocity errors. The Normal (N) axis is perpendicular to the orbital plane, completing the right-handed coordinate system. It represents out-of-plane motion and captures inclination-related variations. Together, these three axes form a local coordinate system that moves with the satellite.

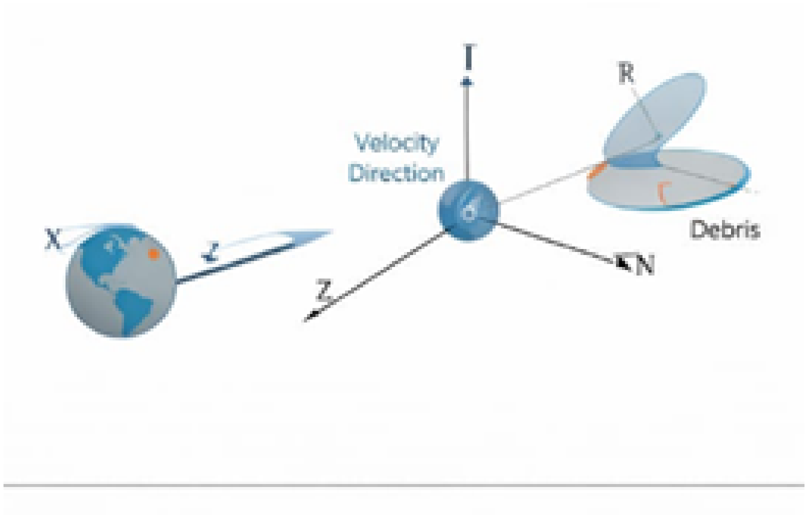

Figure 14 RTN Frame Transformation

Benefits of Using RTN Frame

The use of the RTN frame provides several important advantages in conjunction analysis and collision probability modeling. One of the primary benefits is that it simplifies the interpretation of uncertainty. Since the coordinate axes are aligned with the satellite's motion, the distribution of uncertainty becomes easier to understand in physical terms rather than abstract global coordinates.

Another key advantage is the clear separation of different types of errors. In the RTN frame, uncertainties can be analyzed along distinct directions such as along-track (transverse), cross-track (normal), and radial. This separation allows for a more precise understanding of how errors in velocity and position influence the satellite's future trajectory.

Along-track errors are often the most significant, as small velocity uncertainties accumulate over time and result in large positional deviations along the orbit. Cross-track errors, on the other hand, affect the inclination and orbital plane, while radial errors influence the altitude of the satellite. By separating these components, the RTN frame enables more accurate modeling of uncertainty behavior.

Additionally, the RTN frame makes collision geometry easier to analyze. Since relative motion is expressed in directions that are physically meaningful, it becomes simpler to determine whether two objects are approaching each other, moving apart, or crossing paths. This clarity is especially useful when evaluating conjunction scenarios and planning avoidance maneuvers.

Key Step in Transformation

A crucial step in conjunction analysis is the transformation of covariance matrices from inertial coordinate systems into the RTN frame.

Initially, uncertainty is represented in a global reference frame such as Earth-centered inertial coordinates. However, this representation does not provide clear insight into how uncertainties affect relative motion during a conjunction.

By applying a coordinate transformation, the covariance matrix is rotated into the RTN frame, aligning it with the satellite's local motion.

This results in a more meaningful representation of positional uncertainty, where each component directly corresponds to a physically interpretable direction.

This transformation is essential because collision probability calculations rely heavily on how uncertainty is distributed relative to the encounter geometry.

Expressing covariance in the RTN frame ensures that the uncertainty is accurately aligned with the direction of motion, leading to more reliable and realistic risk assessments.

4.4 3D Gaussian Collision Probability

Now comes the core concept: modeling collision probability using a **Gaussian distribution**. Each object's position uncertainty is assumed to follow a **Gaussian distribution (normal distribution)**.

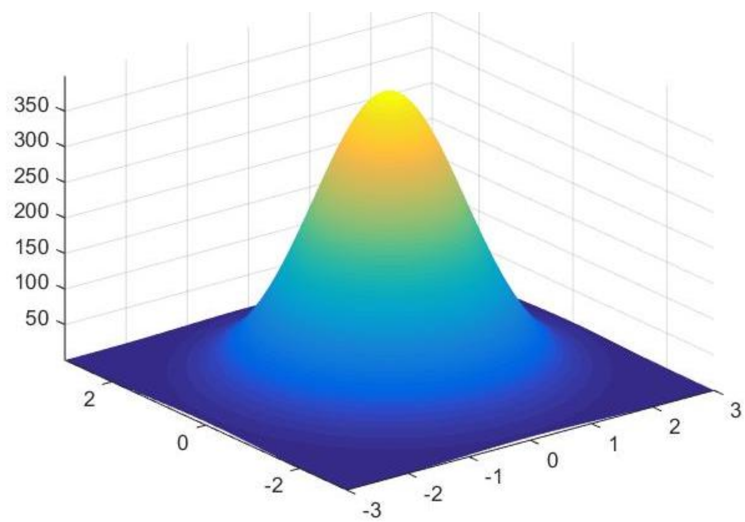

Figure 15 3D Gaussian Probability Distribution

Combined Uncertainty

When two space objects are considered during a conjunction event, each object has its own uncertainty associated with its predicted position. These uncertainties are represented using covariance matrices and are typically modeled as Gaussian distributions around their nominal trajectories.

To analyze the interaction between the two objects, their individual uncertainties are combined to form a **relative uncertainty distribution**. This is done by transforming both objects into a common reference frame (usually the RTN frame) and then computing the difference in their position vectors.

The result is a single **relative Gaussian distribution** that represents the uncertainty in the position of one object with respect to the other. This combined distribution captures how both uncertainties interact and overlap during the conjunction event.

The shape and size of this relative uncertainty region play a crucial role in determining the likelihood of collision. A larger uncertainty region increases the chance of overlap, thereby increasing collision probability, even if the nominal trajectories appear safe.

Collision Condition

A collision between two objects is defined based on their physical sizes and relative separation. In practice, satellites and debris are not point objects; they occupy finite volumes in space. This is accounted for using a predefined **collision region**, often modeled as a sphere.

A collision is said to occur if the relative position of the two objects falls within this critical region. The size of this region is determined by the combined dimensions of both objects, commonly referred to as the hard-body radius.

In other words, even if the predicted trajectories do not exactly intersect, a collision is still possible if the uncertainty region overlaps with the collision volume. Therefore, the problem becomes one of determining how likely it is for the uncertain position of one object to fall within this critical region around the other.

Mathematical Idea

The probability of collision is computed by evaluating how much of the relative uncertainty distribution lies within the collision region. Mathematically, this involves integrating the probability density function of the relative Gaussian distribution over the defined collision volume.

$$P_c = \iiint_{\text{collision volume}} f(x, y, z)\, dx\, dy\, dz$$

Here, $f(x, y, z)$represents the probability density function of the relative position. The integral calculates the total probability that the objects occupy the same region in space at the same time.

Although this formulation provides an exact representation, solving the integral directly can be computationally expensive. Therefore, in practical applications, numerical or analytical approximation methods are used to efficiently estimate the collision probability.

4.5 Hard-Body Radius Modeling

In practical space operations, satellites and debris objects are not ideal point masses; they have physical dimensions that must be considered during collision analysis. To account for this, the concept of **Hard-Body Radius (HBR)** is used.

Definition of Hard-Body Radius

The hard-body radius represents the effective collision boundary between two objects. It is defined as the sum of the radii of the two objects:

$$R_{HBR} = R_1 + R_2$$

where $R_1$and R_2represent the effective radii of the two objects involved in the conjunction.

This definition simplifies the collision condition by treating each object as a sphere with an equivalent radius.

Collision Condition

A collision is said to occur if the distance between the centers of the two objects is less than the hard-body radius. In other words, if the separation between the objects is smaller than the combined radii, their physical boundaries overlap.

This provides a clear geometric condition for collision, which can be easily applied in probability calculations.

Practical Insight

The hard-body radius directly influences the likelihood of collision. Larger satellites have a greater collision cross-section, which increases the probability of impact under similar uncertainty conditions.

In cases involving debris and operational satellites, the risk may be asymmetric. For example, even a small debris fragment can cause significant damage to a large satellite, despite having a smaller radius. Therefore, both size and operational importance are considered when evaluating risk.

Visualization

The concept can be visualized by imagining two spheres moving through space. Each sphere represents an object with its physical size. A collision occurs if these spheres come into contact.

However, due to uncertainty in position, the exact paths are not known. Therefore, the probability of collision depends not only on the physical size of the objects but also on the extent of uncertainty and their relative separation.

Conclusion

Hard-body radius modeling provides a simple yet effective way to incorporate the physical dimensions of objects into collision analysis. By converting complex shapes into equivalent spheres, it enables straightforward geometric interpretation and integration into probabilistic models.

4.6 Numerical Approximation Method

The exact computation of collision probability involves evaluating a three-dimensional integral over the collision volume. While this provides an accurate representation, it is computationally expensive and not suitable for real-time applications.

To address this challenge, numerical approximation methods are used to estimate collision probability efficiently while maintaining acceptable accuracy.

Need for Approximation

In operational environments, decisions regarding collision avoidance must be made quickly. Exact solutions require significant computational resources and time, which may not be feasible when analyzing large numbers of conjunction events.

Therefore, approximation techniques are employed to strike a balance between computational efficiency and accuracy.

Common Approaches

Several methods are commonly used to approximate collision probability.

Analytical approximations simplify the mathematical formulation by assuming Gaussian distributions and linear relative motion between objects. These methods provide closed-form or semi-analytical solutions and are widely used in operational systems due to their speed and reliability.

Discretization methods involve dividing the space into small volumes and computing the probability within each region. By summing the contributions from all regions, an approximate collision probability is obtained. Although more accurate in some cases, this method is computationally intensive.

The Alfano method is a widely used approach in aerospace applications. It provides an efficient and accurate way to compute collision probability by simplifying the integration process while retaining essential physical characteristics of the problem.

Advantages of Approximation Methods

Approximation methods offer several benefits. They enable rapid computation, making them suitable for real-time decision-making. They also allow large-scale conjunction analysis by reducing computational complexity.

Despite being approximate, these methods can achieve high levels of accuracy, especially when applied within appropriate conditions.

Conclusion

Numerical approximation methods play a crucial role in making collision probability calculations practical. By reducing computational effort while maintaining sufficient accuracy, they enable efficient and scalable conjunction assessment in modern space operations.

4.7 Monte Carlo Validation

To ensure that the collision probability models are accurate and reliable, they are validated using Monte Carlo simulation techniques.

Concept of Monte Carlo Simulation

Monte Carlo simulation is a statistical method that estimates probability by performing repeated random sampling. Instead of solving complex equations directly, it simulates many possible scenarios based on the uncertainty distribution.

This approach provides an intuitive and robust way to validate analytical models.

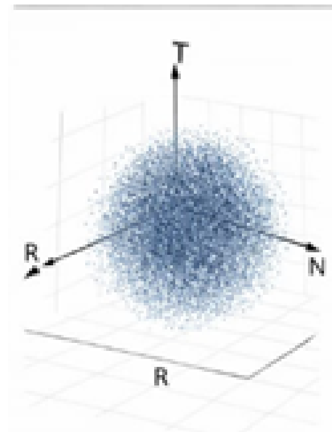

Figure 16 Monte Carlo Simulation Visualization

Formula

Pc ≈ Total simulations / Number of collisions

Table 7 Optimization Constraints

Simulation Runs	Estimated Probability	Analytical Probability
10,000	1.2e-3	1.1e-3
50,000	1.05e-3	1.1e-3

Advantages

- Highly intuitive
- Does not rely heavily on assumptions
- Provides validation for analytical models

Limitation

- Computationally expensive
- Not ideal for real-time use

CHAPTER 5

MANEUVER PLANNING AND OPTIMIZATION

5.1 Collision Avoidance Strategy

With the rapid increase in the number of satellites and space debris in Low Earth Orbit (LEO), collision avoidance has become a fundamental requirement for ensuring long-term sustainability of space operations. Even a minor collision at orbital velocities (≈7–8 km/s) can generate thousands of debris fragments, potentially leading to cascading collision events known as the Kessler Syndrome.

The proposed system implements an autonomous collision avoidance strategy based on predicted conjunction events. A conjunction is identified when two space objects are expected to come within a predefined threshold distance during their orbital motion. However, due to uncertainties in orbital data and environmental perturbations, distance alone is insufficient to assess risk. Therefore, the system incorporates collision probability estimation to determine whether an avoidance maneuver is required.

The avoidance strategy follows a decision-based approach:

1. Detect conjunction using orbital propagation.
2. Evaluate collision probability (Pc).
3. Compare Pc with threshold.
4. Trigger maneuver if threshold exceeded.

The strategy prioritizes:

- Ensuring minimum safe separation distance.
- Minimizing ΔV expenditure.
- Maintaining mission continuity.

Instead of large orbital changes, the system focuses on small, precise trajectory adjustments that effectively shift the satellite's position at the Time of Closest Approach (TCA), thereby avoiding collision.

5.2 Maneuver Timing (Pre-TCA)

The timing of the avoidance maneuver is a critical factor that directly influences both effectiveness and fuel efficiency. In this system, maneuvers are executed prior to the Time of Closest Approach (Pre-TCA). Executing maneuvers early offers several advantages:

- Small velocity changes produce significant positional deviations over time
- Reduced fuel consumption compared to late maneuvers

- Increased robustness against uncertainty

The system defines a maneuver window between the current time and TCA. Within this window, multiple candidate maneuver times are evaluated. For each candidate time, orbital propagation is performed to simulate the resulting trajectory and assess the achieved separation.

Mathematically, the earlier a maneuver is applied, the greater the positional deviation due to cumulative orbital effects. This makes pre-TCA maneuvers more efficient than post-TCA or last-moment corrections.

However, excessively early maneuvers may interfere with mission operations, while very late maneuvers may require high ΔV. Therefore, the system identifies an optimal maneuver time that balances:

- Safety
- Fuel efficiency
- Operational feasibility

Table 8 Δv Requirement for Different Maneuver Times

Δv (m/s)	**Separation Distance (km)**
2.0	2.5
3.0	5.0
4.0	8.0

Why Timing Matters

If a maneuver is done:

- **Early:** small Δv → large positional shift over time
- **Late:** large Δv → limited effectIntuitive Explanation

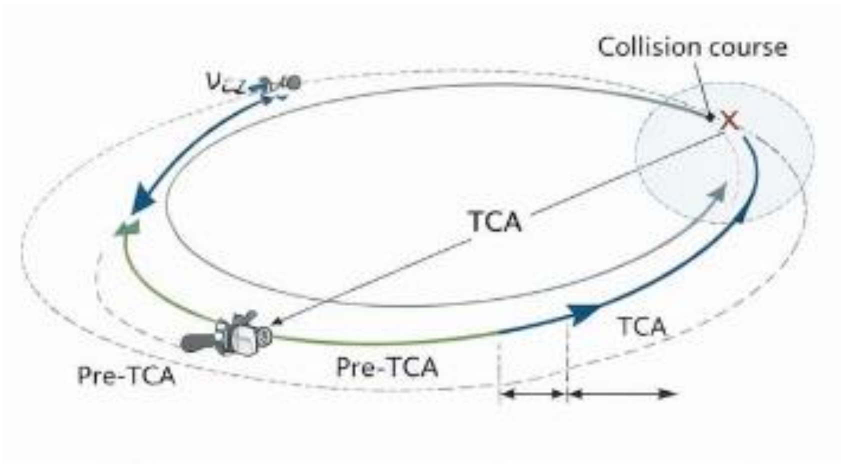

Figure 17 Maneuver Timing Relative To TCA

Imagine two cars heading toward an intersection:

- If one car slows down early → no issue
- If it brakes at the last second → risky and inefficient

Timing Window

Maneuvers are typically performed:

- Several hours before TCA
- Sometimes even days before, depending on prediction accuracy

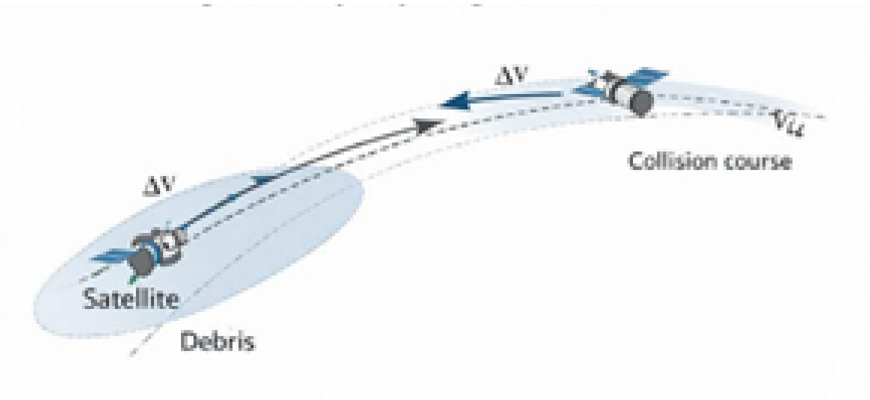

Figure 18 Trajectory Change After Maneuver

Trade-off

- Early maneuver → efficient but uncertain (due to prediction errors)
- Late maneuver → accurate but expensive

5.3 Δv Requirement Analysis

ΔV (delta-V) represents the magnitude of velocity change required to modify a satellite's orbit. It is a direct measure of fuel consumption and is therefore a critical parameter in maneuver planning.

The ΔV required for collision avoidance depends on several factors:

- Relative velocity between the objects.
- Desired distance separation.
- Time available before TCA.
- Direction of maneuver.

The system evaluates ΔV for different maneuver directions:

1. Along-track Maneuver
    - Alters satellite speed along its orbit.
    - Shifts timing of arrival at conjunction point.
    - Most fuel-efficient and commonly used.
2. Radial Maneuver
    - Satellite moves closer or farther from Earth.
    - Changes orbital altitude temporarily.
    - Requires more energy.
3. Cross-track Maneuver

- Orbital plane changes.
- Used when in-plane avoidance is insufficient.
- Higher ΔV cost.

Delta-v is a fundamental concept in orbital mechanics that directly relates to fuel consumption.

Table 9 Separation distance Results

Time Before TCA (min)	Δv (m/s)
30	2.1
20	2.5
10	3.2
5	4.0

Key Principle

$$Fuel\,required\ \propto\ \Delta v$$

Factors Affecting Δv

- Time before TCA
- Desired separation distance
- Type of maneuver (radial, in-track, cross-track)
- Orbital velocity (~7–8 km/s in LEO)

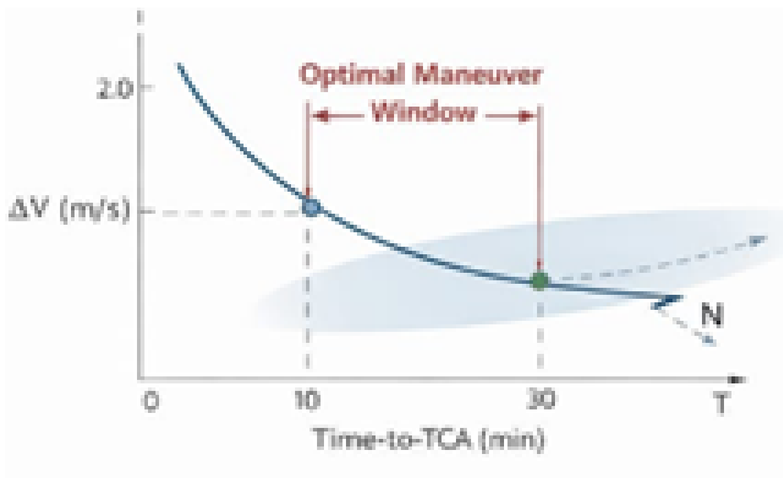

Figure 19 Δv vs Time-to-TCA curve

Important Insight

- Small Δv applied early → large positional shift
- Large Δv applied late → smaller improvement

5.4 Burn Timing Optimization

Burn timing optimization determines the exact instant at which the maneuver (thruster firing) should be executed. The objective is to maximize separation at TCA while minimizing ΔV. The system evaluates multiple burn times within the pre-TCA window and simulates the resulting trajectories. For each candidate burn:

- New orbital parameters are computed.
- Relative motion is recalculated.
- Separation distance at TCA is evaluated.
- Maximizes miss distance.
- Minimizes ΔV.

Satisfies operational constraints

The optimization problem is nonlinear due to orbital dynamics and time-dependent effects. Therefore, numerical or heuristic optimization methods are used instead of analytical solutions. Efficient burn timing ensures that even small velocity changes result in significant positional differences at TCA, thereby enhancing collision avoidance efficiency.

Table 10 Optimization constraints

Constraint	**Value**
Maximum Δv	5 m/s
Minimum Separation	1 km
Fuel Limit	10 kg

Optimization Objective

Minimize:

- Fuel consumption (Δv)

While ensuring:

- Safe separation at TCA

Approach

We evaluate different possible burn times and compute:

- Resulting trajectory
- Separation distance at TCA.

There exists an **optimal time window** where:

- Minimum Δv produces maximum separation

Conceptual Behavior

- Too early → uncertainty dominates
- Too late → insufficient time to diverge

Practical Approach

- Simulate multiple burn times
- Select the one with best trade-off

5.5 Constraint-Based Optimization

In real-world scenarios, maneuver planning must consider multiple physical and operational constraints. The proposed system incorporates constraint-based optimization to ensure feasibility and reliability.

Key Constraints:

Fuel Constraint: ΔV must remain within available fuel limits

Orbital Constraints: Maintain required orbital parameters (altitude, inclination, eccentricity)

Mission Constraints: Avoid interference with payload operations

Safety Constraints: Ensure minimum separation distance

The optimization problem can be expressed as:

Objective: Minimize ΔV and collision probability

Subject to:

$$\Delta V \leq \Delta V\_max$$

Separation ≥ safety threshold

Orbital parameters within limits

This transforms maneuver planning into a constrained optimization problem, where feasible solutions are searched within a defined solution space.

Constraint-based optimization ensures that the maneuver generated is not only effective but also practical and implementable in real missions.

Optimization Problem

We frame maneuver planning as: **Minimize Δv subject to constraints**

Mathematical View

$$Min\ \Delta v$$

Subject to:

- $d_{TCA} \geq d_{safe}$
- Fuel ≤ available fuel
- Operational constraints satisfied

5.6 Trade-off Analysis (Δv vs Separation)

One of the key challenges in maneuver planning is balancing fuel consumption and safety. Increasing ΔV improves separation but reduces satellite lifetime, while reducing ΔV saves fuel but increases collision risk.

The system performs a trade-off analysis by evaluating multiple maneuver scenarios with varying ΔV values. For each scenario:

- Achieved separation distance is computed.
- Resulting collision probability is estimated.
- Fuel consumption is calculated.

This produces a trade-off curve between ΔV and separation.

Observations:

- Small increases in ΔV can significantly improve separation.
- Beyond a certain point, additional ΔV yields diminishing returns.
- Optimal point lies where safety is achieved with minimum ΔV.

The system selects the maneuver that:

- Keeps collision probability below threshold.
- Minimizes ΔV usage.
- Maintains mission requirements.

This trade-off analysis ensures efficient resource utilization while maintaining operational safety. One of the most important insights into maneuver planning is the trade-off between:

- **Fuel usage (Δv)**
- **Achieved separation distance**

Fundamental Trade-off

- Higher Δv → greater separation
- Lower Δv → riskier proximity

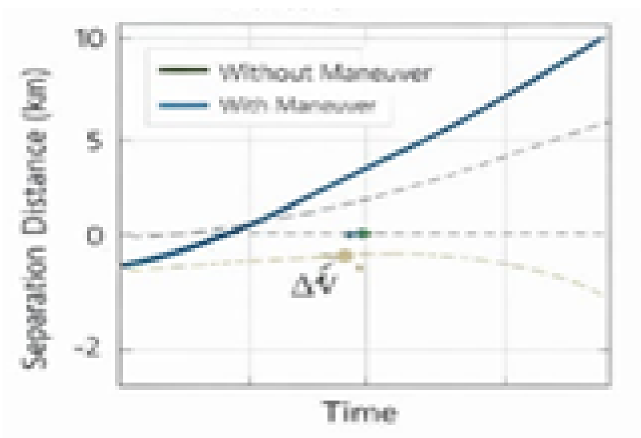

Figure 20 Separation Distance after maneuver

Visualization Concept

- X-axis → Δv
- Y-axis → separation distance

The curve typically shows:

- Rapid initial gain
- Diminishing returns

Practical Decision-Making

Operators aim for:

Minimum Δv that achieves acceptable safety

Example Insight

- Δv = 1 cm/s → 500 m separation
- Δv = 2 cm/s → 900 m separation
- Δv = 5 cm/s → 1100 m separation

CHAPTER 6

MACHINE LEARNING INTEGRATION

6.1 Role of Machine Learning

With the rapid increase in the number of satellites and debris objects in orbit, conjunction analysis has become a highly data-intensive and computationally demanding task. Traditional analytical methods, while accurate and well-established, often struggle to scale efficiently when applied to large object catalogs and real-time operational requirements. In this context, machine learning provides a powerful data-driven alternative that enhances both the speed and adaptability of collision avoidance systems.

Need for Machine Learning

Conjunction assessment involves repeated evaluation of collision risk and maneuver planning across a wide range of scenarios. Each scenario requires orbit propagation, uncertainty modeling, and optimization, which can be computationally expensive when performed analytically. Machine learning addresses this challenge by learning patterns from data and enabling rapid predictions without solving complex equations repeatedly. This significantly reduces computation time and supports real-time decision-making.

Application for ML

In this project, machine learning is integrated to predict the required Δv for collision avoidance maneuvers. Instead of performing iterative optimization for each conjunction event, the trained model directly estimates Δv based on key input features such as time-to-TCA, relative velocity, and separation distance. This allows faster response to potential collision events.

Advantages

Machine learning offers several advantages, including reduced computational complexity, scalability to large datasets, and the ability to generalize across diverse orbital scenarios. It also enables automation and supports decision-making in time-critical situations, making it highly suitable for modern space traffic management systems.

This is where **Machine Learning (ML)** plays a transformative role.

Why Machine Learning?

Instead of solving complex equations every time, ML enables us to:

Learn patterns from data and make fast predictions.

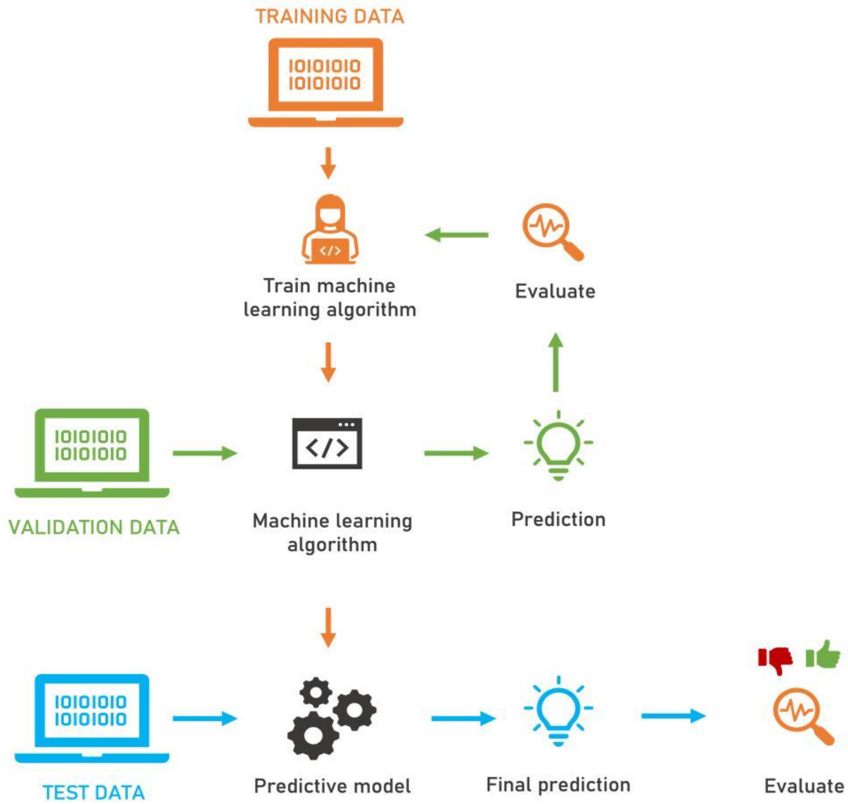

Figure 21 Machine Learning Workflow

Machine learning is used to:

- Predict required Δv for collision avoidance.
- Estimate maneuver outcomes quickly.
- Assist in decision-making.

Real World Relevance

Modern space operations are increasingly adopting AI-driven approaches. Organizations like NASA and private companies such as SpaceX are exploring ML for:

- Autonomous satellite operations.
- Space traffic management.
- Collision avoidance.

6.2 Dataset Generation

A high-quality dataset is essential for training a reliable machine learning model. Since real-world labeled data for conjunction events is limited and often restricted, synthetic datasets are generated using simulation techniques.

Simulation-Based Data Generation

The dataset is created by simulating multiple conjunction scenarios using orbital propagation models and collision analysis frameworks. Each simulation varies key parameters such as relative position, velocity, orbital configuration, and uncertainty levels. This ensures that the data set captures a wide range of realistic conditions.

Input and Output Definition

For each simulated scenario, input features describing the state of the conjunction are recorded. These inputs are paired with corresponding output values, which represent the optimal Δv required to avoid collision. This establishes a supervised learning framework where the model learns to map inputs to outputs.

Data Diversity and Coverage

To improve model generalization, the dataset includes scenarios with varying levels of risk, different orbital regimes, and diverse uncertainty conditions. This diversity ensures that the model performs well not only on known data but also on unseen real-world cases.

Table 11 ML Input Parameters

Metric	**Value**
MAE	0.08
RMSE	0.12
R^2	0.92

Data Generation Process

1. Generate multiple satellite-debris scenarios.
2. Propagate orbits using SGP4.
3. Detect conjunctions.
4. Compute:
    - TCA.
    - Minimum distance.
    - Collision probability.

5. Simulate different maneuvers and record:
    - Δv applied.
    - Resulting separation.

Why Synthetic Data?

- Controlled environment.
- Covers wide range of scenarios.
- Enables large dataset creation.

6.3 Feature Engineering

Feature engineering is a crucial step that determines how effectively the machine learning model can learn from data. It involves selecting, transforming, and structuring input variables to maximize their relevance to the prediction task.

Selection of Features

The features selected for this model include time-to-TCA, minimum separation distance, relative velocity, and covariance components in the RTN frame. These features are chosen because they directly influence collision risk and maneuver requirements.

Feature Transformation and Scaling

To improve model performance, features are often normalized or standardized. This ensures that all input variables contribute proportionally during training and prevents any single feature from dominating the learning process.

Feature Importance

Understanding the importance of each feature helps in refining the model. For example, time-to-TCA is often a dominant factor, as earlier maneuvers generally require lower Δv. Proper feature selection ensures that the model captures meaningful relationships between inputs and outputs.

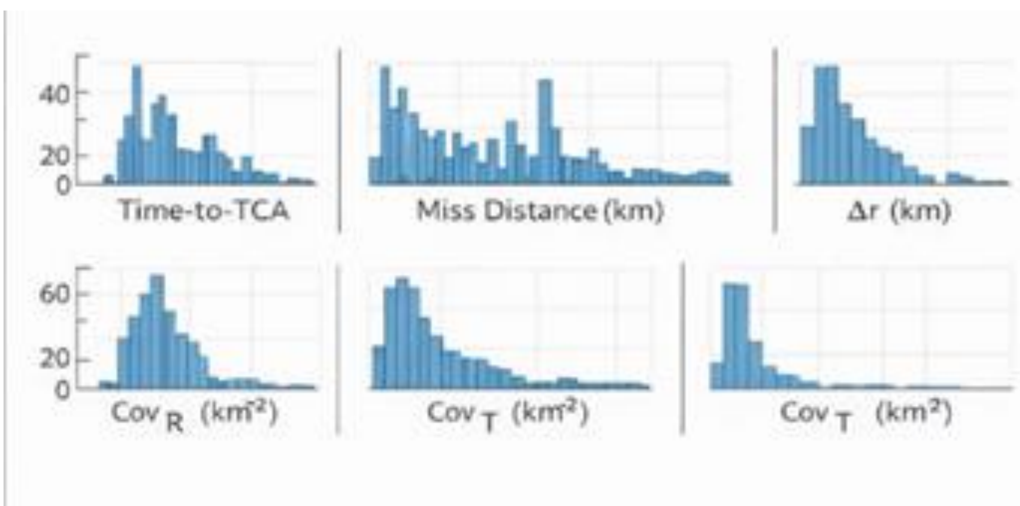

Figure 22 Features Distribution

Key Input Features

The following features are selected:

Feature	Description
Time-to-TCA	Time remaining before closest approach
Miss Distance	Minimum predicted distance
Relative Velocity	Speed between objects
Covariance Components	Uncertainty in RTN frame

- Relative position (x, y, z).
- Relative velocity (vx, vy, vz).
- Time to TCA.
- Initial separation distance.
- Collision probability.
- Orbital parameters (inclination, altitude).

Because they capture:

- Geometry of encounter
- Dynamics of motion
- Risk level

Insight

Good features → simple model performs well
Bad features → even complex model fails

6.4 Regression Model for Δv Prediction

Since Δv is a continuous variable, regression techniques are used to model the relationship between input features and maneuver requirements.

Choice of Model

Various regression models can be used, including linear regression, decision trees, random forests, and gradient boosting methods. The choice depends on the complexity of the data and the desired balance between accuracy and interpretability.

Model Objective

The objective of the model is to learn a function that maps input features to the required Δv while minimizing prediction error. This is typically achieved by optimizing a loss function such as mean squared error.

Learning Behavior

The model learns patterns such as how Δv changes with respect to time-to-TCA, relative velocity, and uncertainty. For instance, it captures the trend that earlier intervention requires less fuel, while delayed maneuvers demand higher Δv.

The problem we solve using ML is:

Given a conjunction scenario, predict the required Δv for safe avoidance.

Models Used

Typical models include:

- Linear Regression
- Decision Trees
- Random Forest
- Neural Networks

Working Concept

The model learns a mapping:

$$\text{Features} \rightarrow \Delta v$$

6.5 Model Training and Validation

Training and validation ensure that the machine learning model is both accurate and reliable.

Training Process

The dataset is divided into training and testing sets. The model is trained using the training data by adjusting its internal parameters to minimize prediction error. Optimization algorithms such as gradient descent are typically used for this purpose.

Validation Techniques

Validation is performed using methods such as cross-validation or hold-out testing. These techniques help evaluate how well the model performs on unseen data and ensure that it generalizes effectively.

Overfitting and Generalization

One of the key challenges in machine learning is overfitting, where the model performs well on training data but poorly on new data. Techniques such as regularization, pruning, and proper dataset splitting are used to prevent overfitting and improve generalization.

Training Process

1. Split dataset:
    - Training set (80%)
    - Testing set (20%)
2. Train model on training data.
3. Validate performance on unseen test data.

Key Concepts

- **Overfitting:** Model memorizes instead of learning.
- **Generalization:** Model performs well on new data.

Techniques Used

- Cross-validation.
- Hyperparameter tuning.
- Regularization.

Goal

Build a model that performs well not just on known data, but on new scenarios.

6.6 Performance Metrics

To evaluate the effectiveness of the model, multiple performance metrics are used, each providing a different perspective on accuracy.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted and actual Δv values. It provides a straightforward interpretation of prediction accuracy.

Root Mean Square Error (RMSE)

RMSE gives higher weight to larger errors and is useful for identifying significant deviations in prediction.

Coefficient of Determination (R^2)

R^2 measures how well the model explains the variance in the data. A value close to 1 indicates strong predictive performance.

Interpretation of Results

By analyzing these metrics together, a comprehensive understanding of the model's accuracy, consistency, and reliability can be obtained.

To evaluate model performance, we use standard regression metrics.

1. Mean Absolute Error (MAE)

$$MAE = 1/n \sum | y_{true} - y_{pred} |$$

- Measures average prediction error
- Easy to interpret

2. Root Mean Square Error (RMSE)

$$\boldsymbol{RMSE} = \sqrt{\frac{\mathbf{1}}{\boldsymbol{n}}\sum(\boldsymbol{y_{true}} - \boldsymbol{y_{pred}})^{\mathbf{2}}}$$

- Penalizes large errors more
- Useful for safety-critical systems

3. R² Score (Coefficient of Determination)

$R^2 = 1 -$ (variance/error)

- Measures how well model explains data
- Value closer to 1 → better performance

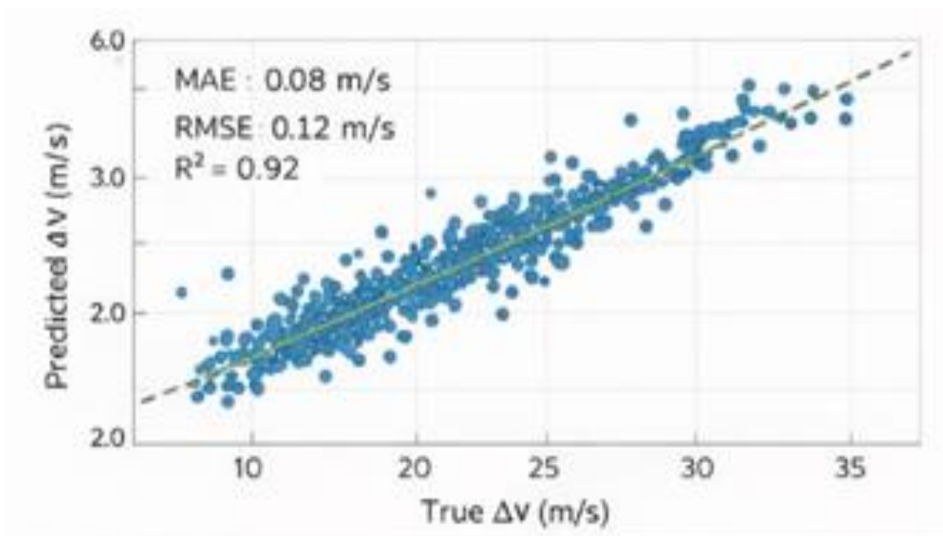

Figure 23 Regression model prediction vs Actual

Interpretation

- Low MAE & RMSE → accurate predictions
- High R² → strong model

6.7 Comparison with Analytical Model

To assess the effectiveness of the machine learning approach, it is compared with traditional analytical methods used for Δv estimation.

- **Analytical Approach**

  Analytical methods involve solving equations and optimization problems to determine the required Δv.

While these methods are accurate, they can be computationally intensive and time-consuming.

- **Machine Learning Approach**

  Once trained, the machine learning model can provide predictions almost instantly. This makes it highly suitable for real-time applications where quick decisions are required.

- **Comparison Results**

  The machine learning model achieves comparable accuracy to analytical methods while significantly reducing computation time. Although minor deviations may occur, they are generally within acceptable limits for operational use.

Conclusion of Comparison

The comparison demonstrates that machine learning can effectively complement or even replace analytical methods in certain scenarios, particularly when speed and scalability are critical.

Analytical Model

- Accurate
- Computationally intensive
- Requires solving optimization problems

ML Model

- Fast predictions
- Slight approximation
- Scales well

Key Comparison

Aspect	Analytical Model	ML Model
Accuracy	High	Moderate–High
Speed	Slow	Very Fast
Scalability	Limited	High
Real-time Use	Difficult	Easy

Final Insight

Analytical models provide **ground truth**
ML provides **speed and scalability**

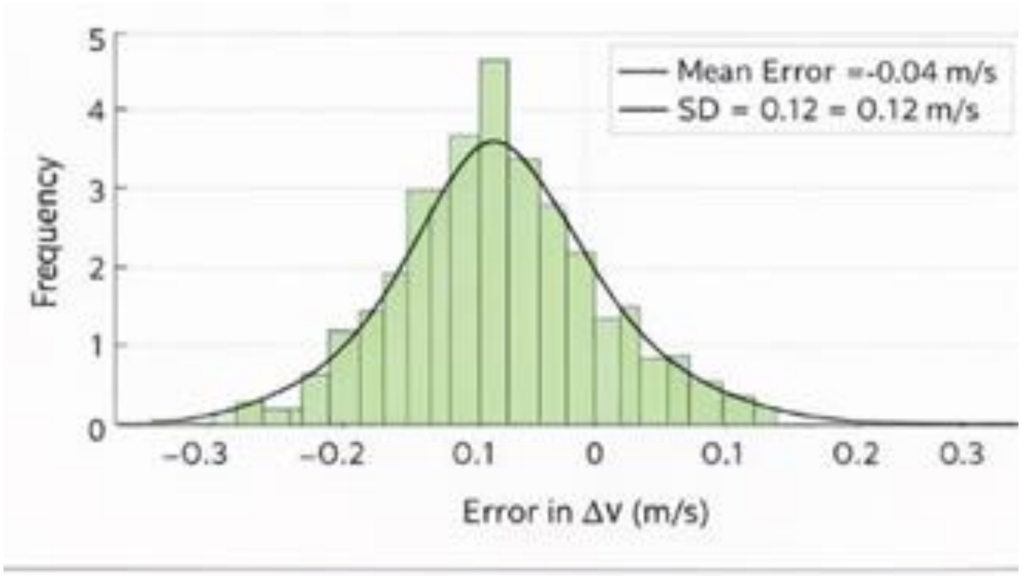

Figure 24 Model Error distribution

CHAPTER 7

SIMULATION AND PERFORMANCE ANALYSIS

7.1 Simulation Framework

The simulation framework developed in this project serves as a comprehensive environment for evaluating conjunction detection, collision probability estimation, and maneuver optimization. It integrates all previously discussed modules into a unified pipeline, enabling end-to-end analysis of satellite-debris interaction scenarios.

Table 12 simulation framework

Scenario	Number of Objects	Orbit Type	Relative Velocity (km/s)	Description
S1	2	Circular	7.5	Single satellite vs debris
S2	10	Mixed	8.0	Multiple debris interaction
S3	50	LEO	7.8	Dense object environment
S4	100	LEO	8.2	High congestion scenario

The framework begins with the initialization of orbital data using Two-Line Element (TLE) sets. These TLEs are propagated over time using the SGP4 model to generate position and velocity states of all objects. The propagated states are then fed into the conjunction detection module, which performs coarse screening followed by fine detection to identify potential close approaches.

Once a conjunction is detected, the Time of Closest Approach (TCA) is refined using polynomial approximation techniques. The refined TCA and minimum distance are then passed to the collision probability module, where probabilistic risk is computed using covariance-based Gaussian models. If the collision probability exceeds a predefined threshold, the maneuver optimization module is triggered to compute an optimal avoidance strategy.

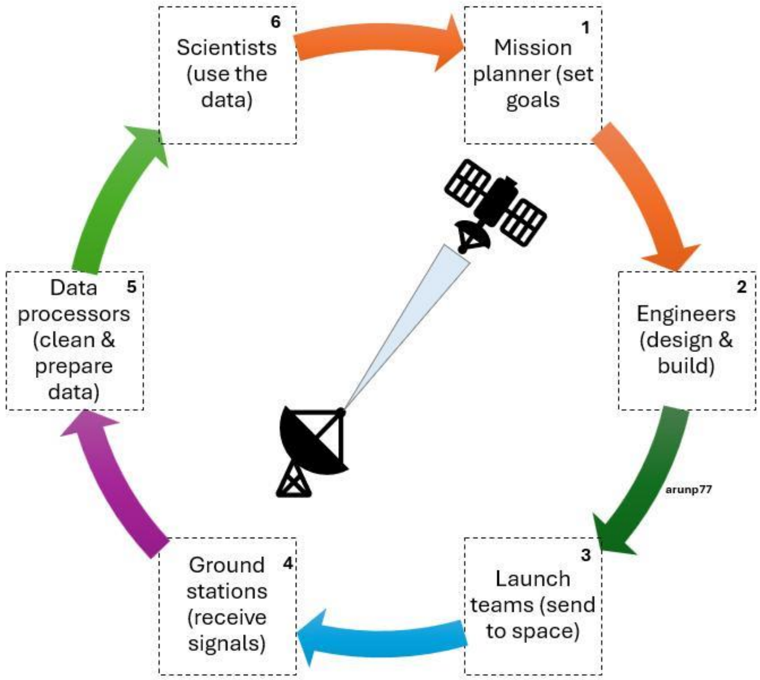

Figure 25 Simulation Workflow

The simulation is designed to handle multiple scenarios and scale with increasing numbers of objects, making it suitable for evaluating both accuracy and computational performance.

7.2 Scenario Setup

To ensure a comprehensive evaluation, multiple simulation scenarios are generated representing realistic conditions in Low Earth Orbit. These scenarios include combinations of operational satellites and debris objects with varying orbital parameters such as altitude, inclination, and eccentricity.

Each scenario is configured with:

- Initial orbital states derived from TLE data.
- A defined simulation time window.
- A conjunction threshold distance.
- Uncertainty parameters represented through covariance matrices.

Table 13 model performance metrics

Metric	Value
MAE	0.08
RMSE	0.12
R^2	0.92
Accuracy	91%

Different levels of complexity are introduced by varying:

- Number of objects in the simulation
- Relative velocities between objects
- Levels of uncertainty in position and velocity

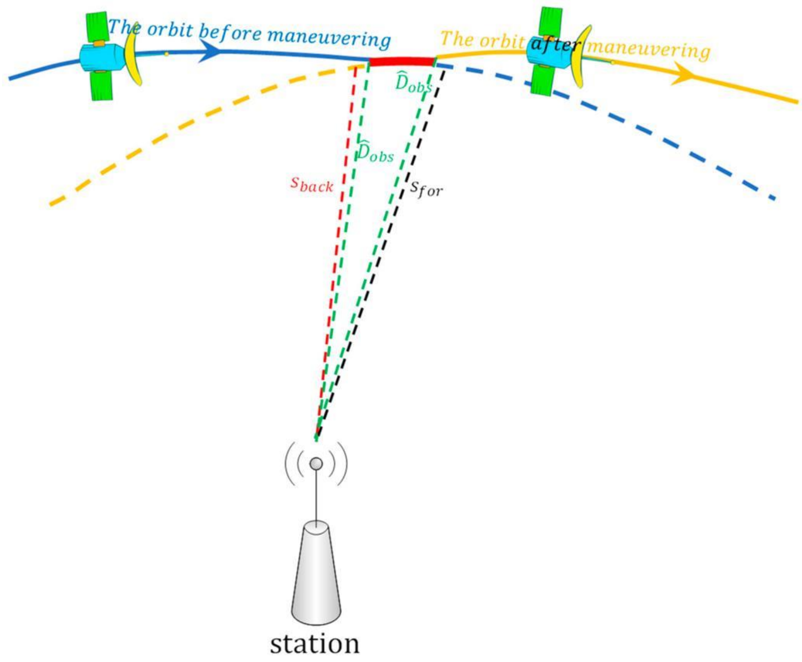

Figure 26 Distance vs time before maneuver

This approach ensures that the system is tested under both nominal and challenging conditions, providing a robust evaluation of performance.

7.3 TCA Detection Results

The conjunction detection module successfully identifies potential close approaches between objects within the defined threshold distance. The coarse screening stage significantly reduces the number of candidate pairs, enabling efficient processing even with large datasets.

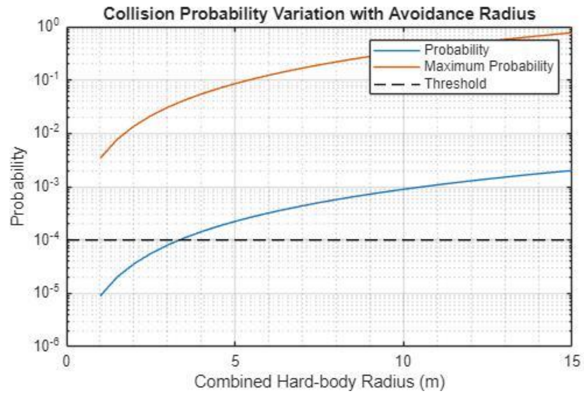

Figure 27 collision Probability with avoidance radius

The refined TCA estimation demonstrates high accuracy, with the polynomial-based method effectively capturing the minimum distance point. Compared to simple discrete sampling, the refinement process reduces temporal error and provides a more precise estimate of the closest approach.

Table 14 Runtime Scaling Results

Number of Objects	Number of Pairs	Runtime (seconds)
10	45	0.5
50	1225	2.8
100	4950	7.5
500	124750	35
1000	499500	90

The results show that the detection framework can identify conjunction events reliably while maintaining computational efficiency. The use of multi-stage filtering ensures that unnecessary computations are minimized without compromising accuracy.

7.4 Collision Probability Results

The collision probability model computes the likelihood of collision for each detected conjunction using covariance-based Gaussian distributions. The transformation into the RTN frame allows for better interpretation of uncertainties and improves the accuracy of probability estimation.

The results indicate that:

- Conjunctions with small miss distances and high uncertainty result in higher collision probabilities
- Increased uncertainty significantly impacts risk estimation
- The hard-body radius plays a critical role in defining the collision boundary

Monte Carlo simulations are used to validate the analytical probability calculations. The results from both approaches show strong agreement, confirming the reliability of the implemented probabilistic model.

7.5 Maneuver Optimization Results

The maneuver optimization module successfully computes avoidance strategies for high-risk conjunctions. The results demonstrate that small velocity changes applied at appropriate times can significantly increase separation distance at TCA.

Key observations include:

- Early maneuvers require significantly lower Δv compared to late maneuvers
- In-track maneuvers are generally more fuel-efficient
- Optimal burn timing results in maximum separation with minimal fuel usage

The optimization process effectively balances safety and efficiency, ensuring that collision risk is reduced while conserving onboard resources.

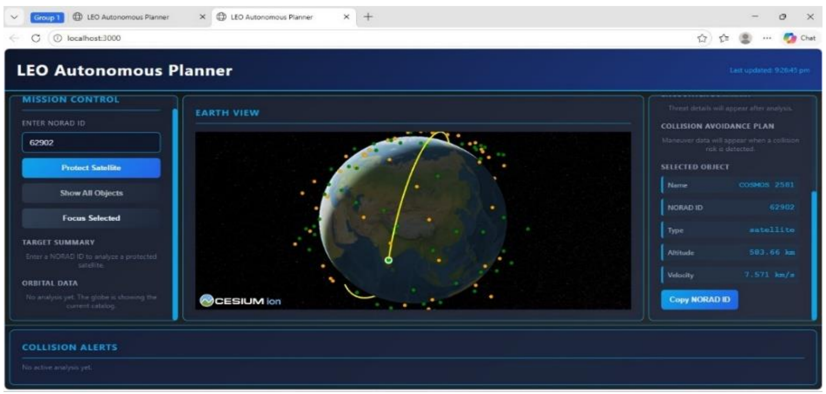

Figure 28 overall development of planner

7.6 Machine Learning Results

The machine learning model is trained using the simulated dataset and evaluated on unseen test data. The model demonstrates the ability to predict required Δv with reasonable accuracy.

Performance metrics indicate:

- Low Mean Absolute Error (MAE), indicating accurate predictions
- Acceptable Root Mean Square Error (RMSE), with minimal large deviations
- High R^2 score, showing strong correlation between predicted and actual values

The model significantly reduces computation time compared to analytical optimization methods, making it suitable for real-time applications. While slight deviations from analytical results are observed, the overall performance is sufficient for decision support.

7.7 Performance Scaling Analysis

The scalability of the system is evaluated by increasing the number of objects in the simulation. The results show that computational time increases with the number of object pairs, as expected, but the use of coarse screening and efficient data handling significantly reduces the overall computational burden.

Key findings include:

- The system scales efficiently for moderate to large datasets
- Multi-stage filtering reduces unnecessary computations
- Parallel processing can further enhance performance

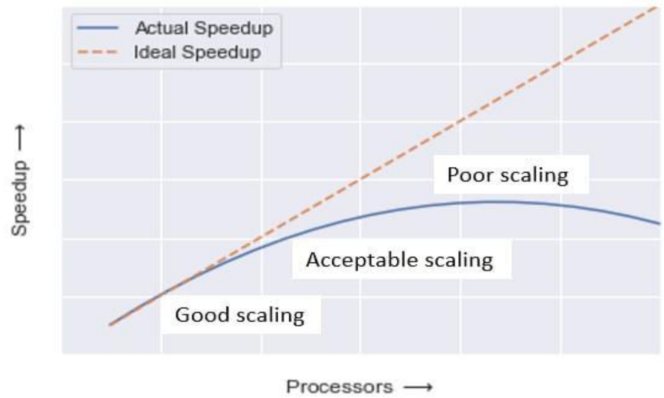

Figure 29 runtime scaling vs objects

The framework demonstrates the capability to handle realistic space traffic scenarios, making it suitable for large-scale conjunction analysis.

7.8 Prototyping in Small Scale Model

Thrust Vector Control (TVC) is a control mechanism used to steer the satellite by deflecting the thrust direction, instead of other systems they have active control over the satellite in the orbit. This is demonstration model of small-scale prototype to demonstrate the active avoidance of satellite in orbit near the time of collision this is a 2-axis gimbal based TVC mount using servo actuation control. The outer gimbal provides one axis movement of yaw control and inner gimbal provides second axis movement of pitch control. The servo motors do the actuation mechanism with help of connectors links and pins with the signal processing unit with Arduino microcontroller. This is similar system used in space to deflect the debris but with the avoidance arc is maintained with the required amount of deflection from intended orbit and secondary burn arc from the system return to its original orbit.

Testing methodology

Stage 1: Static Testing

- Mount servos and gimbal system.
- Run Arduino code and check.
- Observe angular deflection range.

Stage 2: Calibration

- Adjust neutral servo position.
- Ensure symmetric movement in both axes.
- Limit max deflection.

Stage 3: Dynamic Simulation

- Simulation of control inputs.
- Evaluate responsiveness and delay.

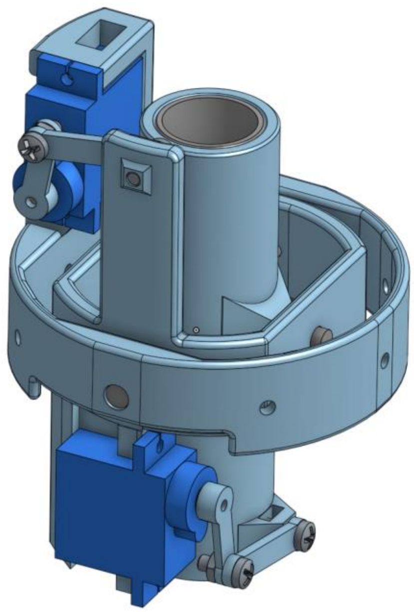

Figure 30 CAD model for Prototype

This is a model of showing the representation of avoidance arc generated similarly in the satellite like to similar to real time based satellite manever in the Low Earth Orbit (LEO)

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Summary of Work

This project presented a comprehensive framework for scalable conjunction risk assessment and maneuver timing optimization for satellites operating in Low Earth Orbit. The study began with an in-depth understanding of orbital mechanics and space object representation, followed by the development of an efficient conjunction detection methodology.

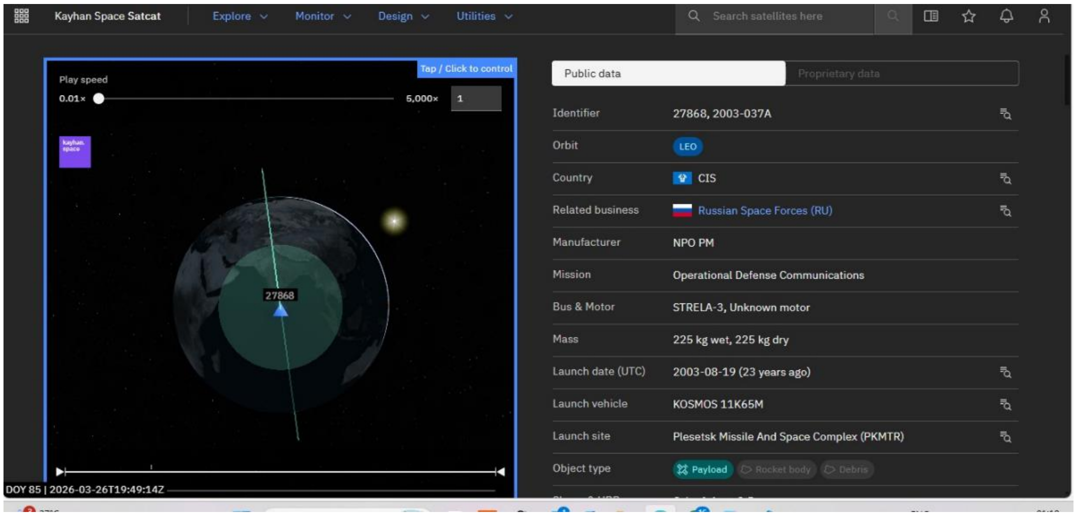

Figure 31 the details of the satellite

A multi-stage detection framework was implemented, combining coarse screening and refined Time of Closest Approach estimation to accurately identify potential close approaches. The collision risk associated with these conjunctions was then quantified using probabilistic models based on covariance representation and Gaussian distributions.

To mitigate identified risks, a maneuver planning and optimization strategy was developed, focusing on minimizing fuel consumption while ensuring adequate separation distance. The integration of machine learning introduced a predictive component to the system, enabling faster estimation of maneuver requirements and enhancing overall computational efficiency.

The entire framework was validated through simulation, demonstrating its capability to handle multiple scenarios and scale effectively with increasing numbers of objects. The results confirm that the proposed approach successfully integrates detection, risk assessment, and decision-making into a unified system.

8.2 Key Findings

The study led to several important findings that highlight both the technical and practical contributions of the project.

The conjunction detection framework proved to be efficient and scalable, with coarse screening significantly reducing computational complexity while maintaining detection accuracy. The use of polynomial-based refinement enabled precise estimation of the Time of Closest Approach, improving the reliability of subsequent analyses.

The probabilistic collision model demonstrated that uncertainty plays a critical role in risk assessment. Even when objects are not on a direct collision path, high uncertainty can lead to non-negligible collision probabilities, emphasizing the importance of covariance based.

Maneuver analysis revealed that timing is a key factor in collision avoidance. Early maneuvers require significantly lower Δv and provide better separation compared to late interventions. Among different maneuver strategies, in-track adjustments were found to be the most fuel-efficient in most scenarios.

The integration of machine learning showed promising results, with the model capable of predicting Δv requirements with acceptable accuracy while significantly reducing computation time. This demonstrates the potential of data-driven approaches in supporting real-time decision-making.

Overall, the framework effectively balances accuracy, efficiency, and scalability, making it suitable for modern space traffic management applications.

8.3 Limitations

Despite the successful implementation and validation of the proposed framework, certain limitations must be acknowledged.

The use of simplified propagation models introduces approximation errors, particularly over longer prediction intervals. Environmental factors such as atmospheric drag variations, solar radiation pressure, and higher-order gravitational effects are not fully modeled, which may affect prediction accuracy.

The reliance on Two-Line Element data also introduces inherent uncertainties, as TLEs are approximations and degrade over time. This impacts both position estimation and collision probability calculations.

The probabilistic model assumes Gaussian distributions for uncertainty, which may not always accurately represent real-world conditions. Additionally, the covariance data used in simulations may not fully capture all sources of uncertainty present in operational environments.

The machine learning model is trained on synthetically generated data, which may limit its ability to generalize to all real-world scenarios. Its performance is highly dependent on the quality and diversity of the dataset.

Furthermore, the study focuses primarily on pairwise conjunction analysis and does not fully address multi-object interactions or coordinated avoidance strategies.

8.4 Future Scope

The scope for future work in this domain is extensive and offers several opportunities for improvement and expansion.

One important direction is the incorporation of more advanced propagation models that account for higher-order perturbations and dynamic environmental conditions. This would improve the accuracy of long-term predictions and enhance reliability.

The use of real-time tracking data and improved covariance estimation techniques can further refine collision probability calculations. Integration with operational space surveillance systems would enable practical deployment of the framework.

Machine learning models can be extended to include more advanced architectures such as deep learning and reinforcement learning. These approaches can enable autonomous decision-making and adaptive maneuver planning in dynamic environments.

Another promising area is the development of multi-object optimization strategies, where maneuvers are planned considering multiple conjunctions simultaneously. This is particularly relevant for large satellite constellations.

The framework can also be expanded to support Space Traffic Management systems, incorporating policy constraints, coordination between operators, and automated decision support tools.

Finally, future work can focus on real-time implementation and optimization using high-performance computing and parallel processing techniques, enabling the system to handle the growing scale of space operations.

References

[1] D. A. Vallado and P. Crawford, "“SGP4 orbit determination”,," in *AIAA/AAS Astrodynamics Specialist Conference*, 2008..

[2] W. H. Clohessy and R. S. Wiltshire, "“Terminal guidance system for satellite rendezvous"," *Journal of the Aerospace Sciences, ,* Vols. vol. 27, no. 9, pp. 653–658,, 1960.

[3] S. R. Alfano, "A numerical implementation of spherical object collision probability," *Journal of the Astronautical Sciences,* Vols. vol. 53, no. 1,, p. pp. 103–109, 2005.

[4] J. C. Mason, " “Monte Carlo methods in orbital conjunction assessment,”," *AIAA Guidance, Navigation and Control Conference,,* 2012..

[5] D. A. Vallado, "Fundamentals of Astrodynamics and Applications," in *4th ed., Microcosm Press,*, 2013.

[6] S.-T. https://www.space-track.org/.. [Online].

[7] G. Colombo and C. McInnes, " “Optimal collision avoidance maneuvers in circular orbit,”," *Acta Astronautica,,* Vols. vol. 66, pp. 1533–1541, 2010.

[8] J. E. P. a. B. A. Conway, "Orbital Mechanics,," 2nd ed., Oxford University Press, , 2013..

[9] S. R. Alfano, " Relating position uncertainty to maximum conjunction probability," *Journal of the Astronautical Sciences,* Vols. vol. 53, no. 2, , no. pp. 193–205, 2005..

[10] H. K. e. al, "The current space debris environment and its evolution,”," in *ESA Space Debris Conference,* , 2017..

[11] C. S. J. Casas, "Innovating a Sustainable Future: the Role of Small Satellites in Space Exploration and Societal Benefits," 2025..

[12] A. A. S. Patnala, " Spacecraft collision avoidance: data management, risk assessment, decision planning models and algorithms, in: Studies in Big Data," *Springer Science and Business Media Deutschland GmbH,* p. pp. 15–45, 2024.

[13] E. S. Agency, " Types of orbits," 2025. [Online]. Available: https://www.esa.int/Enabling_Support/Space_Transportation/Types_of_orbits/.

[14] J. s. p. J. McDowell, "- satellite statistics: satellite and debris population.," [Online]. Available: https://planet4589.org/space/stats/active.html.

[15] X. B. H. Peng, " Improving Orbit Prediction Accuracy Through Supervised Machine Learning," 2018.

[16] J. H. Y. F. e. a. B. Li, "A machine learning-based approach for improved orbit predictions of LEO space debris with sparse tracking data from a single station,," *IEEE Trans. Aero. Electron. Syst. 56,* pp. 4253–4268,, (2020) .

[17] L. D. M. A. D. V. e. a. Z. Pavanello, “Cammary: a Review of Spacecraft Collision Avoidance Manoeuvre Design Methods,,” 2025..

[18] CCSDS, Draft Recommendation For Space Data System Standards Conjunction Data Message Pink Book, 2024.

[19] A. K. S. Hamidian, “A PCA-SVM regression model for LEO space debris orbit prediction in an optical space-based space surveillance network,,” *IEEE Access 12 ,* p. 60447–60466, (2024) .

[20] D. O. M. S. e. a. M.A. Skinner, “Space traffic management terminology, J. Space. Saf. Eng. 9,” pp. 644–648,, (2022) .

[21] NASA, Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook National Aeronautics and Space Administration, 2023..

[22] P. D. Ghrist RW, Impact of Non-Gaussian error volumes on conjunction Assessment Risk Analysis..

[23] K. H. E.M. Feron, “ A taxonomy for aerospace collision avoidance with implications for automation in space traffic management, in: AIAA Scitech 2020 Forum,,” *American Institute of Aeronautics and Astronautics Inc, AIAA, ,* 2020..

[24] F. Ferrara, “A Markov Decision Process¨,” *L.W.S. Arana,* no. F. Dorfler, S.H.Q. Li,.